

**PERSONALIZED IMMUNOTHERAPY IN SEPSIS: A MULTICENTRE AND
MULTINATIONAL, DOUBLE-BLIND, DOUBLE-DUMMY RANDOMIZED CLINICAL
TRIAL (THE IMMUNOSEP TRIAL)**

STUDY PROTOCOL

Authors:

Evangelos J. Giamarellos-Bourboulis, MD, PhD

Mihai G. Netea, MD, PhD

Sponsor and CRO:

Hellenic Institute for the Study of Sepsis

88 Michalakopoulou str., 11528 Athens, Greece

insepsis@otenet.gr

tel: 0030 210 7480662

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The undersigned legal representatives of the Hellenic Institute for the Study of Sepsis, we confirm that the version 1.0 dated 4 December 2020 of the protocol of the ImmunoSep trial confirms to the current guidelines current International Council for Harmonisation (ICH) Guideline for Good Clinical Practice (GCP).

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LIST OF ABBREVIATIONS

AC	acute cholangitis
AE	adverse event
aPPT	activated partial thromboplastin time
BSI	primary bacteremia
CAP	community-acquired pneumonia
COVI-9	Coronavirus disease 2019
DIC	disseminated intravascular coagulation
DMC	Data Monitoring Committee
DSUR	drug safety updated report
EC	Ethics Committees
eCRF	electronic Case Report Form
EDTA	ethylene-diamene-tetracetic acid
GCP	Good Clinical Practice
G-CSF	granulocyte-colony stimulating factor
HAP	hospital-acquired pneumonia
HBD	hepatobiliary dysfunction
HISS	Hellenic Institute for the Study of Sepsis
HIV	human immunodeficiency virus
HLA-DR	histocompatibility complex DR
HSSG	Hellenic Sepsis Study Group
IB	investigator's brochure
ICF	Informed Consent Form
IL	Interleukin
IMP	investigational medicinal product
INR	international normalized ratio

IRB	Institution Review Boards
ISF	Investigator Site File
IV	Intravenous
LPS	lipopolysaccharide
MALS	macrophage activation-like syndrome
MAP	mean arterial pressure
MDSCs	myeloid-derived suppressor cells
OR	odds ratio
PBMCs	peripheral blood mononuclear cells
qSOFA	quick SOFA
RCT	randomized clinical trial
rhIFN γ	recombinant human activated interferon-gamma
ROC	receiver operator characteristics
SAE	serious adverse event
SARS-CoV-2	Severe acute respiratory syndrome coronavirus-2
Sc	subcutaneous
SIRS	systemic inflammatory response syndrome
SIV	site initiation visit
SMB	Safety Monitoring Board
SOFA	sequential organ failure assessment
STEAE	serious treatment-emergent adverse events
SUSAR	suspected, unexpected serious adverse reaction
TB	Tuberculosis
TEAE	treatment-emergent adverse events
TNF α	tumor necrosis factor-alpha
VAP	ventilator-associated pneumonia

SYNOPSIS

Aim	Previous findings coming from the randomized clinical trial PROVIDE have showed that serum ferritin and expression of HLA-DR on circulating monocytes are reliable diagnostic markers of macrophage activation syndrome and of sepsis-induced immunoparalysis. The aim of ImmunoSep is to use these biomarkers and to study using a randomized design if adjunctive immunotherapy targeting either fulminant hyper-inflammation or immunoparalysis is able to improve sepsis outcomes.
Design	Prospective, multicentered, interventional, randomized, double-blind, double-dummy, placebo-controlled clinical trial
Inclusion criteria	• Age equal to or above 18 years • Male or female gender • In case of women, unwillingness to become pregnant during the study period • Written informed consent provided by the patient or by their legal representative in case of patients unable to consent • Community-acquired pneumonia (CAP) or hospital-acquired pneumonia (HAP) or ventilator-associated pneumonia (VAP) or primary bacteremia (BSI) • Sepsis defined by the Sepsis-3 definitions • Patients with signs of fulminant hyper-inflammation or sepsis-associated immunoparalysis; patients diagnosed with COVID-19 may participate only in the fulminant hyper-inflammation arm • Time from classification into sepsis by the Sepsis-3 definitions and start of blind intervention less than 72 hours
Study groups	• Standard-of-care therapy Patients will receive standard-of-care treatment decided by the attending physicians. They will also receive 20ml intravenous (IV) 0.9% saline (N/S) three times daily (every eight hours) for 15 days and 0.5 ml subcutaneous (sc) 0.9% N/S every other day for a total of 15 days. • Immunotherapy Patients will receive standard-of-care treatment decided by the attending physicians. They will also receive IV anakinra 200 mg three times daily (every eight hours) or sc rhIFNγ 100 μg once every other day. More precisely, patients randomized for hyper-inflammation will receive anakinra three times daily (every eight hours) for 15 days and sc 0.5 ml N/S 0.9% every other day for 15 days. Patients having immunoparalysis will

	receive IV 20ml N/S 0.9% three times daily (every eight hours) for 15 days and sc rhIFNγ every other day for 15 days. Especially for patients with creatinine clearance lower than 30 ml/min anakinra will be given half dose (i.e. 100 mg three times daily/10 ml 0.9% saline). Creatinine clearance is calculated by the Cockcroft Gault equation $[(140 - \text{age in years}) / (72 \times \text{serum creatinine in mg/dl})]$ for men; this is multiplied by 0.85 for women.
Primary study endpoint	The study is aiming to the effect of a strategy of personalized immunotherapy on morbidity in patients with sepsis or septic shock and sepsis-associated hyperinflammation or immunoparalysis. The primary efficacy endpoint is the difference in the mean total SOFA (Sequential Organ Failure Assessment) score until day 9 after randomization.
Secondary study endpoint	• 28-day mortality • 90-day mortality • The change of mean total SOFA score on day 15 of the end of treatment • The impact of personalized immunotherapy on the reversal of hyperinflammation or immunoparalysis. • The impact of personalized immunotherapy on the resolution of infection leading to study enrolment. • The development of genomic, epigenomic, proteomic, metabolomic and microbiomic surrogate biomarkers for the primary and secondary endpoints. This will come from the exploitation of the genomic and proteomic material that will be analyzed by the partners of the ImmunoSep project.
Power of the study	The study is powered for the primary endpoint, i.e. decrease of SOFA score by at least 1.4 points on day 9. Considering a drop-out rate of about 15%, a total of 280 patients need to be randomized.

BACKGROUND

Sepsis is a life-threatening organ dysfunction that results from the dysregulated host response to an infection¹. Accumulating knowledge suggests that there is a spectrum of dysregulation in this response. On the one end of this spectrum there are patients whose immune response is characterized by fulminant hyper-inflammation. On the other end of this spectrum there are patients whose immune response is characterized by immunoparalysis. The majority of patients are situated between these two extremes. The primary hypothesis of the ImmunoSep trial is to recognize both ends of this spectrum and to administer adjunctive therapy aiming to modulate the sepsis-associated hyper-inflammation or immunoparalysis. It is anticipated that with this strategy patients' organ dysfunctions be improved.

During the last years the Hellenic Sepsis Study Group (HSSG) managed to develop ferritin as the diagnostic tool for the recognition of patients with fulminant sepsis-associated hyper-inflammation. This was done by analysis of 5,121 patients split into a test and a validation cohort and by also studying a confirmation cohort coming from Sweden. Patients were classified according to the criteria for the macrophage activation syndrome developed by the American College of Rheumatology; approximately 4% of patients with sepsis have fulminant hyper-inflammation or macrophage activation-like syndrome (MALS) that is an independent clinical condition associated with short-term 10-day mortality². Serum ferritin greater than 4,420 ng/ml had sensitivity 97.1% and negative predictive value 98% for the diagnosis. More than 25 years ago one randomized clinical trial (RCT) was conducted where patients with severe sepsis were randomly assigned to blind treatment with placebo or with the recombinant human interleukin-1 receptor antagonist anakinra³. The trial failed to disclose any benefit of anakinra on 28-day mortality. However, a recent *post-hoc* analysis revealed that patients who had signs of macrophage activation syndrome had significant 30% survival benefit by anakinra treatment⁴.

The immunoparalysis of sepsis is associated with at least 50% risk of death in the subsequent 28 days. There is evidence from preclinical studies and from the endotoxin challenge model in human volunteers that this can be reversed using recombinant human interferon gamma (rhIFN γ)⁵. rhIFN γ was administered in nine

patients with septic shock in a small open-label clinical trial without placebo comparator; reversal of immunoparalysis was achieved⁶.

From these data we can conclude that it is important to recognize patients with sepsis complicated either with MALS or with immunoparalysis and administer anakinra or rhIFN γ respectively as a potentially beneficial intervention. To this end, we conducted a smaller-scale trial in Greece that was aiming to the personalized management of septic shock. The acronym of this trial was PROVIDE (see Final Clinical Study report in APPENDIX I). PROVIDE was conducted between December 2017 and December 2019 in 14 study sites in Greece under the auspices of the European Shock Society. In the PROVIDE trial patients with septic shock due to lower respiratory tract infection, acute cholangitis, or primary bacteremia, were screened on two consecutive days for laboratory signs of fulminant hyper-inflammation or immunoparalysis. Results showed that one single measurement of serum ferritin and the number of human leukocyte antigen-DR (HLA-DR) on monocytes can efficiently classify patients. More precisely, ferritin $\geq 4,420$ ng/ml diagnoses MALS; and a combination of ferritin $\leq 4,420$ ng/ml and HLA-DR less than 5000 molecules/monocyte diagnose immunoparalysis. Patients were randomized into double-dummy blind treatment with placebo if randomly assigned to the standard-of-care arm and with anakinra/recombinant human interferon-gamma (rhIFN γ) if randomly assigned to the immunotherapy arm. Thirty-six patients were enrolled and preliminary results derived from the PROVIDE trial further corroborate the use of anakinra/rhIFN γ as an innovative, personalized adjunct therapy for sepsis but one larger-scale study with larger number of patients is needed in order to validate findings.

ImmunoSep is a randomized placebo-controlled phase 2 clinical trial with a double-dummy design where the effect of personalized immunotherapy in patients with sepsis and either fulminant hyper-inflammation or immunoparalysis is studied. We consider hyper-inflammation as a more direct life-threatening manifestation of sepsis than immunoparalysis; for that reason patients with lab findings of both immune states are allocated to the hyper-inflammation treatment arm.

AIM OF THE STUDY

Our aim is to conduct one RCT in sepsis to assess whether personalized adjunctive immunotherapy directed against a state of either fulminant hyper-

inflammation or immunoparalysis is able to change sepsis outcomes. Patients will be selected by a panel of biomarkers and laboratory findings and will be allocated to placebo or immunotherapy treatment according to their needs. The study enrolment will be competitive between the participating study sites.

STUDY DESIGN

This will be a prospective randomized placebo-controlled phase 2 clinical trial in a total number of 24 study sites. These study sites are recorded at Appendix II.

This clinical study falls under Directive 2001/20/EC (Clinical Trials Directive). The study protocol will be submitted for approval to the Ethics Committees (EC) and Institution Review Boards (IRB) as per participating study site regulations. For countries where national authority approval is needed, the protocol will be submitted to the central regulatory (e.g. National Ethics Committee of Greece and National Organization for Medicines of Greece). The study will be provided one EudraCT number and will be registered at the EudraCT public site of the EU (<https://eudract.ema.europa.eu>). Following approval from the ECs and IRBs and from the central regulatory bodies, each study site will be initiated. The study will also be registered at ClinicalTrials.gov (www.clinicaltrials.gov) before the initiation of the first study site.

The study shall be governed by:

- The international standards for Good Clinical Practice (GCP) developed by the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use
- Directive 2001/20/EC for Clinical trials
- General Data Protection Regulation 679/2016 (EC)

Study population

Patients should meet ALL the following criteria in order to be enrolled:

Inclusion criteria

- Age equal to or above 18 years.
- Both genders.
- In case of women, unwillingness to become pregnant during the study period.

- Written informed consent provided by the patient or by one first-degree relative/spouse in case of patients unable to consent.
- Community-acquired pneumonia (CAP) or hospital-acquired pneumonia (HAP) or ventilator-associated pneumonia (VAP) or primary bacteremia (BSI). These are defined in APPENDIX III and IV using three sets of criteria per infection. Enrolled patients should meet criteria as defined in APPENDIX III.
- Sepsis defined by the Sepsis-3 definitions. More precisely, sepsis is defined as the presence of total SOFA (sequential organ failure assessment score) equal to 2 or more for patients who are admitted with infection at the emergency department OR as any increase of admission SOFA by 2 or more points for patients already hospitalized¹. The calculation of the SOFA score is provided in APPENDIX V.
- Patients with signs of fulminant hyper-inflammation or sepsis-associated immunoparalysis as defined in APPENDIX VI. Since the state of hyper-inflammation is considered more life-threatening than the state of immunoparalysis, patients with lab findings of both immune states are allocated to treatment targeting hyper-inflammation. It is explicitly stated that patients diagnosed with COVID-19 infection may participate only in the fulminant hyper-inflammation arm
- Time from classification into sepsis by the Sepsis-3 definitions and start of blind intervention less than 72 hours.

Patients meeting ANY of the following criteria CANNOT be enrolled in the study:

Exclusion criteria

- Age below 18 years.
- Denial for written informed consent.
- Acute pyelonephritis or intraabdominal infection, meningitis or skin infection.
- Any stage IV malignancy.
- Neutropenia defined as an absolute neutrophil count lower than 1,500/mm³.
- Any 'do not resuscitate' decision in the hospital.

- In the case of BSI, patients with blood cultures growing coagulase-negative staphylococci or skin commensals or catheter-related infections cannot be enrolled.
- Active tuberculosis (TB) as defined by the co-administration of drugs for the treatment of TB.
- Infection by the human immunodeficiency virus (HIV).
- Any primary immunodeficiency.
- Oral or intravenous intake of corticosteroids at a daily dose equal or greater than 0.4 mg/kg prednisone or greater the last 15 days.
- Any anti-cytokine biological treatment the last one month.
- Medical history of systemic lupus erythematosus.
- Medical history of multiple sclerosis or any other demyelinating disorder.
- Pregnancy or lactation. Women of child-bearing potential will be screened by a urine pregnancy test before inclusion in the study.

Screening for eligibility

Patients eligible for the study are patients either admitted to hospital from the emergency department or patients already hospitalized in the general ward or in an intensive care unit. Once a patient is presenting with at least two of the signs of the systemic inflammatory response syndrome (SIRS) or at least one sign of the quick SOFA, then he/she or a legal representative in case the patient cannot consent, are been asked for written informed consent. Screening for eligibility can only start after written informed consent is provided by the patient or legal representative. Once the written informed consent is provided, then the patient is screened for eligibility using the following screening steps. No study related procedure will be performed prior to obtaining written informed consent.

- Step 1: The patient is screened for the exclusion criteria. If he/she meets any of them, the patient cannot be enrolled. If the patient does not meet any of them, he/she remains eligible and screening proceeds to step 2.
- Step 2: The patient is screened for the underlying infection using already available information from his/her file. This information comprises thorough history, vital signs, physical examination, white blood cell counting, biochemistry, blood gasses, microbiology, chest X-ray and abdominal ultrasound and computed

tomography. None of these examinations will be performed for the needs of the ImmunoSep study. If the patient does not meet the criteria for any of the infections listed in APPENDIX III, he/she cannot be enrolled. If he/she meets the criteria for only one of the infections defined in APPENDIX III, the patient remains eligible and screening proceeds to step 3. Screening at that stage also includes RT-PCR of one nasopharyngeal swab for SARS-CoV-2

- Step 3: The patient is screened for sepsis once all laboratory findings are available. Classification into sepsis or not is done using the SOFA score as described in the inclusion criteria. If the patient is classified positive for sepsis, he/she remains eligible and screening proceeds to step 4. The exact timepoint when the patient is classified to have sepsis will be recorded.
- Step 4: This is the most crucial step in the screening procedure since this is where the classification of the immune state of the patient as fulminant hyper-inflammation or sepsis-associated immunoparalysis is taking place. As already described for the PROVIDE study, patients can be classified into three immunogroups: (a) those with predominant hyper-inflammation with serum ferritin >4,420 ng/ml irrespective of Quantibrite and who are at a state of immunoparalysis; (b) those with Quantibrite below 5,000 AB/C and serum ferritin below 4,420 ng/ml; (c) and those who are unclassified. Patients of immunogroups (a) and (b) may be enrolled in the trial provided they meet the time difference of less than 72 hours from classification into sepsis.

These suggested tools will be used in step 4 of screening of patients for the ImmunoSep trial using the following daily procedure: 6 ml of whole blood is drawn after venipuncture of one forearm vein under aseptic conditions. The blood is poured into two vials; 3ml into one pyrogen-and anticoagulant free tube that is centrifuged for serum isolation; and another 2 ml into one EDTA-coated tube. Following each sampling, tubes are transported to the central lab. The central lab will be at close proximity to the study site for patients enrolled in Lyon (France), Jena (Germany), Amsterdam and Nijmegen (the Netherlands), Cluj-Napoca (Romania) and Lausanne (Switzerland). For study sites in Greece, tubes will be transported via courier on the same day to the central lab that is the Research Laboratory of Immunology of Infections of the 4th Department of Internal Medicine at ATTIKON University Hospital. Upon arrival at the lab, ferritin is measured by an enzyme immunosorbent assay

within six hours. The tube containing EDTA-blood is used for flow cytometry of the expression of HLA-DR molecules on CD14/CD45 positive cells using the Quantibrite assay.

The lab needs to report early to the un-blinded pharmacist investigators of each study site the values of serum ferritin and HLA-DR on monocytes. The website that is already set up by the HISS will be used (www.sepsis.gr). More precisely, two un-blinded pharmacist investigators (one main and a backup investigator) are delegated at each study site. Once the lab report is ready, they receive an e-mail informing that the report is ready. Then by using a username and a password generated for them, they log into the site to retrieve the result.

The overall screening process is shown in Figure 1.

The pharmacists will fill the provided values of ferritin and Quantibrite of the screening samples on the screening webpage. They will then immediately receive the feedback if the patient can be enrolled in the study and to what intervention the patient is allocated. The timepoint when this web-allocation is done is recorded. **The patient can be enrolled in the study only if the difference between this timepoint and the timepoint of sepsis classification designated at screening step 3 is less than 72 hours apart.**

All collected data coming from screening visits will be uploaded to the electronic Case Report Form (eCRF) system of HISS. The platform is safe and it is based on the availability of a personal server (www.sepsis.gr). Each principal investigator and each sub-investigator who remain blinded to the allocated study drug will be provided a unique username and password allowing uploading of data to the study eCRF platform. Each patient who is added in the platform as “newly screened” will be provided with a code containing a letter and a 3-digit number. The letter refers to the study site and the 3-digit number to the serial number of screening. In case the screened patient is enrolled, he maintains this code.

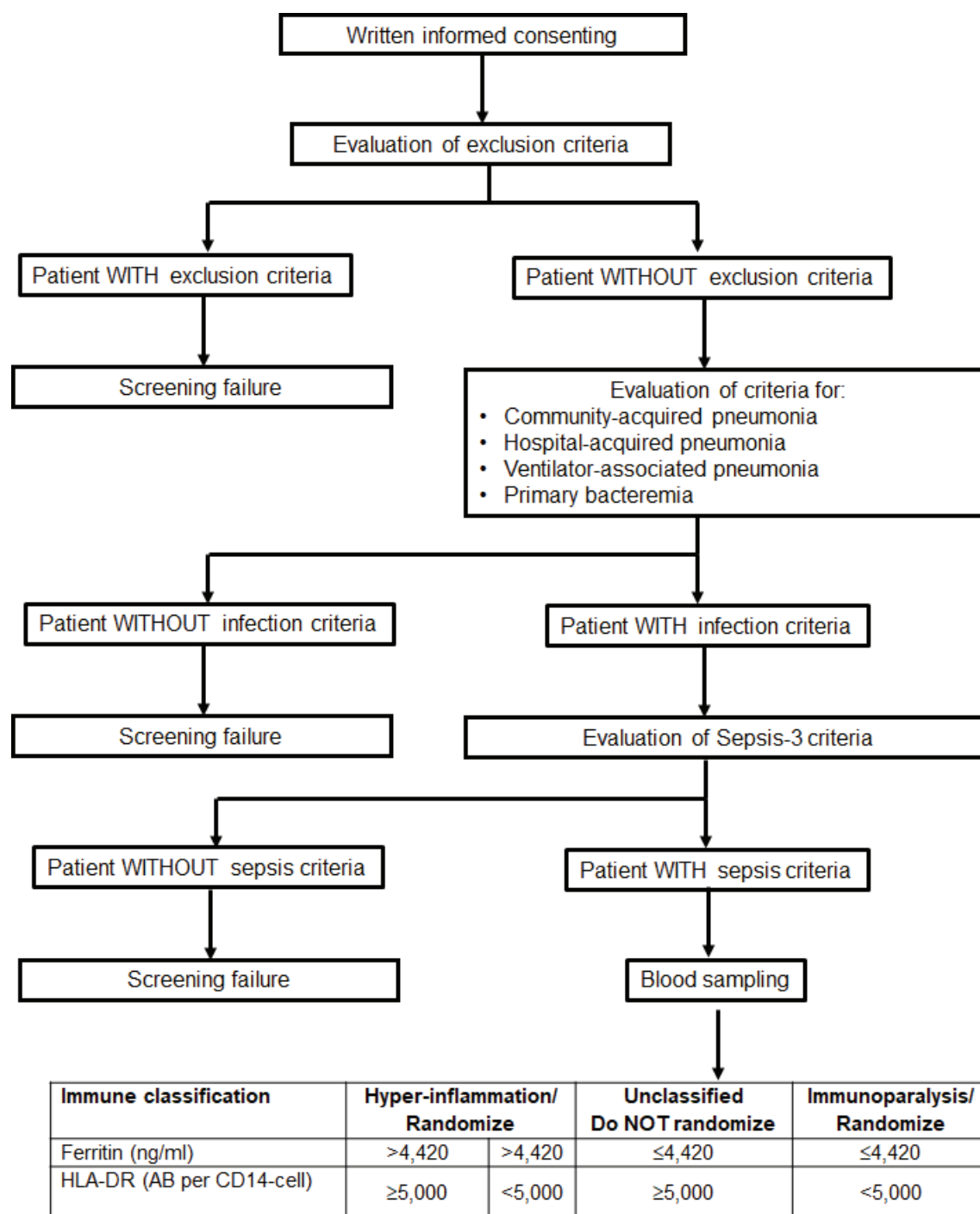


Figure 1: Screening process for patient eligibility for enrolment in the ImmunoSep study

Allocation to blind treatment

Once a patient is considered eligible for enrolment, he will be blindly allocated at 1:1 ratio to one of the following groups of treatment. A separate randomization list will be generated for each study site by the Sponsor. The preparation and the administration of the study drug will be done in each study site by the un-blinded pharmacist investigators. The two groups of treatment are as follows:

- **Standard-of-care therapy** Patients will receive the standard type of treatment decided by the attending physicians. They will also receive 20ml (10ml for patients with creatinine clearance lower than 30ml/min) intravenous (IV) 0.9% saline (N/S) three times daily (every eight hours) for 15 days and 0.5 ml subcutaneous (sc) 1ml 0.9% N/S every other day for a total of 15 days.
- **Immunotherapy** Patients will receive the standard type of treatment decided by the attending physicians. They will also receive IV anakinra 200 mg three times daily (every eight hours) or sc rhIFN γ 100 μ g once every other day. More precisely, patients randomized for hyper-inflammation will receive anakinra three times daily (every eight hours) for 15 days and sc 0.5 ml N/S 0.9% every other day for 15 days. Patients having immunoparalysis will receive IV 20 ml N/S 0.9% (10ml for patients with creatinine clearance lower than 30ml/min) three times daily (every eight hours) for 15 days and sc rhIFN γ every other day for 15 days. Especially for patients with creatinine clearance lower than 30 ml/min anakinra will be given half dose (i.e. 100 mg three times daily). Creatinine clearance is calculated by the Cockcroft Gault equation $[(140 - \text{age in years}) / (72 \times \text{serum creatinine in mg/dl})]$ for men; this is multiplied by 0.85 for women.

The total period of 15 days of treatment is decided using the survival curves of overall screened patients of the PROVIDE study. According to this study, a considerable drop in the survival of patients with hyper-inflammation and immunoparalysis is taking place until day 15 after which the survival rate remains constant until day 28.

Study drug

The active study drug i.e. anakinra and rhIFN γ and placebo will be provided to each site in the form of pre-filled ready-to-use syringes. All study drugs need to be stored at 2-8°C at the study site at a refrigerator with recording of temperature. In case recording indicates deviation of temperature below 0°C or above 10°C for more

than a day, stored drugs need to be replaced. The un-blinded site pharmacists are responsible for the recording of the storage conditions of the study drug. They also need to prepare the test article according to instructions provided at investigator's brochure (IB). The IB will be provided by the Sponsor. Syringes with active drug or placebo will be covered to conceal the identity of the test article. Covering materials will be provided by the Sponsor. At the exterior of each syringe after coverage, a label will be placed with a letter and a 6-digit number. The letter refers to the study site, the first two digits of the number refer to the serial number of the enrolled patient at the respective study site, the middle two digits of the number refer to the day of treatment and the last two digits of the number refer to the time of treatment. For example, the code A010202 refers to study site A, patient number 01 at that study site, treatment day 2 and second injection for that day. The un-blinded pharmacist will provide the covered syringes to the blinded nurse or blinded investigator who will administer the infusion.

Patients' visits and interventions (APPENDIX VII)

Clinical data will be collected on study visits. The collection of data can be done only by the principal investigators and the sub-investigators who are blind to the allocated treatment. The following will be done on each study visit as follows:

Day 1 (Visit 1)

This visit will take place on the day of the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-morbidities, co-administered drugs, past-history, SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Blood sampling to be processed as stated below
- Collection of microbiome samples from skin, nares, oral cavity and rectum using swabs
- Administration of study drug (recording of the exact time of administration)
- Capture of any TEAE or STEAE

Day 2 (Visit 2)

This visit will take place on the morning of the second day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Blood sampling to be processed as stated below; this is done before the administration of the first dose of the investigational drugs of this day
- Capture of any TEAE or STEAE
- Administration of study drug (recording of the exact time of administration)

Day 3 (Visit 3)

This visit will take place on the morning of the third day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Capture of any TEAE or STEAE
- Administration of study drug (recording of the exact time of administration)

Day 4 (Visit 4)

This visit will take place on the morning of the fourth day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Capture of any TEAE or STEAE

- Blood sampling to be processed, as stated below; this is done before the administration of the first dose of the investigational drugs of this day
- Collection of microbiome samples from skin, nares, oral cavity and rectum using swabs; this is done before the administration of the first dose of the investigational drugs of this day
- Administration of study drug (recording of the exact time of administration)

Day 5 (Visit 5)

This visit will take place on the morning of the fifth day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Capture of any TEAE or STEAE
- Administration of study drug (recording of the exact time of administration)

Day 6 (Visit 6)

This visit will take place on the morning of the sixth day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Recording of the clinical state of the infection making the patient eligible for the study
- Capture of any TEAE or STEAE
- Administration of study drug (recording of the exact time of administration)

Day 7 (Visit 7)

This visit will take place on the morning of the seventh day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Recording of the clinical state of the infection making the patient eligible for the study
- Capture of any TEAE or STEAE
- Blood sampling to be processed as stated below; this is done before the administration of the first dose of the investigational drugs of this day
- Collection of microbiome samples from skin, nares, oral cavity and rectum using swabs; this is done before the administration of the first dose of the investigational drugs of this day
- Administration of study drug (recording of the exact time of administration)

Day 8 (Visit 8)

This visit will take place on the morning of the eighth day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Recording of the clinical state of the infection making the patient eligible for the study
- Capture of any TEAE or STEAE
- Administration of study drug (recording of the exact time of administration)

Day 9-14 (Visit 9-14)

This visit will take place on the morning of the 9th – 14th day from the start of treatment with the study drug. The following procedures will be done on that days in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available);

biochemistry (if available); coagulation (if available); blood gasses (if available),
microbiology and antibiogram (if available)

- Recording of the clinical state of the infection making the patient eligible for the study
- Capture of any TEAE or STEAE
- Administration of study drug (recording of the exact time of administration)

Day 15 (Visit 15)

This visit will take place on the morning of the 15th day from the start of treatment with the study drug. This is also the last day of the administration of the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)
- Administration of study drug (recording of the exact time of administration)
- Blood sampling to be processed as stated below; this is done after the administration of the last dose of the investigational drugs of this day
- Collection of microbiome samples from skin, nares, oral cavity and rectum using swabs; this is done after the administration of the last dose of the investigational drugs of this day
- Recording of the clinical state of the infection making the patient eligible for the study
- Capture of any TEAE or STEAE

Day 21 (Visit 16)

This visit will take place on the morning of the 21st day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Recording of co-administered drugs; SOFA score; vital signs; absolute blood cell count and differentiation, hemoglobin and absolute platelet count (if available); biochemistry (if available); coagulation (if available); blood gasses (if available), microbiology and antibiogram (if available)

- Blood sampling to be processed as stated below
- Collection of microbiome samples from skin, nares, oral cavity and rectum using swabs
- Recording of the clinical state of the infection making the patient eligible for the study
- Capture of any TEAE or STEAE

Day 28 (Visit 17)

This visit will take place on the morning of the 28th day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Capture of any TEAE or STEAE
- Recording of survival
- Blood sampling to be processed as stated below
- Collection of microbiome samples from skin, nares, oral cavity and rectum using swabs

In case the patient has deceased, the information will rely on the patient file. In case the patient is discharged, this visit will be done by phone call.

Day 90 (Visit 18, last visit)

This visit will take place on the morning of the 90th day from the start of treatment with the study drug. The following procedures will be done on that day in hierarchical order:

- Capture of any AE or SAE
- Recording of survival
- Blood sampling to be processed as stated below
- Collection of microbiome samples from skin, nares, oral cavity and rectum using swabs

In case the patient has deceased, the information will rely on the patient file. In case the patient is discharged, this visit will be done by phone call.

The clinical state of the infection on any of the evaluable days will be considered as⁷:

- Resolved, according to the criteria for infection resolution provided in APPENDIX III
- Failure, if infection persists or relapses
- Superinfection, if a new infection due to a new pathogen emerges
- Intermediate, if the outcome cannot be classified in one of the above categories

Laboratory procedures

A total of 31.5 ml of blood is collected under puncture of one antecubital vein under aseptic conditions on the following study visits:

- Day 1 (visit 1) This is collected before the administration of the study drug so as to correspond to the baseline condition
- Day 2 (visit 2), one day after trial drug administration
- Day 4 (visit 4)
- Day 7 (visit 7)
- Day 15 (visit 15)
- Day 21 (visit 16)
- Day 28 (visit 17)
- Day 90 (visit 18, last visit)

The collected blood is aliquoted and processed as follows:

- a) 2.5ml into one PAXgene tube to be used for transcriptomic analysis. PAXgene tubes are stored refrigerated at -80°C;
- b) 3 ml into a first EDTA-coated tube (ethyldiamine tetraecetic acid) to be used for flow cytometry
- c) 3 ml into a second EDTA-coated tube for the isolation of plasma; Plasma is stored refrigerated at -80°C for metabolomics analysis pending;
- d) 10 ml into a third EDTA-coated tube for the isolation of peripheral blood mononuclear cells (PBMCs) and their ex vivo stimulation for cytokine production.
- e) 10 ml into a fourth EDTA-coated tube. CD14-monocytes and CD4-lymphocytes will be isolated by differential centrifugation and purified by negative selection using magnetic beads. They will be stored refrigerated at -

80°C pending RNA-sequencing and ATAC-epigenome sequencing.

Transcriptome and epigenome will be performed on days 1, 4, 7, 28 and 90.

- f) 3ml into one sterile and pyrogen-free tube to be used for the isolation of serum. Ferritin will be measured in serum by an enzyme immunosorbent assay. The remaining serum will be stored refrigerated at -80°C for proteomic analysis pending.

At the exterior of each tube a label will be placed with a letter and a 5-digit number. The letter refers to the study site, the first three digits of the number refer to the serial number of the enrolled patient at the respective study site and the last two digits of the number refer to the day of sampling. For example, the code A00104 refers to study site A, patient number 001 at that study site and sampling day 4. This type of labelling allows maintaining patient anonymity. Following each sampling, tubes are transported to the central lab. The central lab will be at close proximity to the study site for patients enrolled in Lyon (France), Jena (Germany), Amsterdam and Nijmegen (the Netherlands), Cluj (Romania) and Lausanne (Switzerland). For study sites in Greece, tubes will be transported via courier on the same day to the central lab that is the Research Laboratory of Immunology of Infections of the 4th Department of Internal Medicine at ATTIKON University Hospital. This mode of transportation via courier in Greece has already been set up and working for the ongoing PROVIDE study.

Swabs of microbiome samples arriving in each lab will be labeled in a way similar to the labeling of blood processed aliquots. They will be stored refrigerated at -80°C pending analysis.

STUDY ENDPOINTS

The primary efficacy study endpoint will be the comparative difference in the mean total SOFA (Sequential Organ Failure Assessment) score until day 9 after randomization. At least 1.4 points decrease of the mean total SOFA score in the immunotherapy arm compared to the standard-of-care arm on day 9 of follow-up must be achieved. For patients dying before day 9, their mean SOFA score until the day of death will be used for this comparison.

The secondary study endpoints will be the efficacy of the applied immunotherapy versus standard of care on each of the following regarding difference of the two groups on:

- 28-day mortality
- 90-day mortality
- The change of mean total SOFA score on day 15 of the end of treatment
- The impact of personalized immunotherapy on the reversal of hyper-inflammation or immunoparalysis. This endpoint applies on day 15 of the end of treatment and it is defined as follows: a) for patients with fulminant hyper-inflammation as any at least 15% decrease of the baseline serum ferritin; and b) for patients with sepsis-associated immunoparalysis as restoration of Quantibrite to above 8,000 AB/C with serum ferritin below 4,420 ng/ml

Exploratory study endpoints will be:

- The impact of personalized immunotherapy on the resolution of infection leading to study enrolment. This endpoint applies on day 15 and the criteria for the resolution of each infection are provided in APPENDIX I.
- The development of genomic, epigenomic, proteomic, metabolomic and microbiomic surrogate biomarkers for the primary and secondary endpoints. This will come from the exploitation of the genomic and proteomic material that will be analyzed by the partners of the ImmunoSep project.

POWER OF THE STUDY

The study is powered for the primary endpoint, i.e. decrease of SOFA score by at least 1.4 points on day 9. In order to calculate the power of the study the following hypotheses are made:

- According to data from the previous randomized clinical trials in sepsis VISEP⁸ and MAXSEP⁹ on 1,137 patients in total, there is a significant association between the mean total SOFA score at day 9 after randomization (primary endpoint of the present study) and 28-day mortality. This is also shown by the follow-up of patients screened for enrolment in the PROVIDE study. A reduction of the primary endpoint by 1.4 points is expected to be associated with a reduction of 28-day mortality.

- Based on the preliminary results of the PROVIDE study, 40% of patients of the enrolled patients in each arm will be recruited with fulminant hyper-inflammation and another 60% for sepsis-associated immunoparalysis.
- The study is powered for 90% at the 5% level of significance.
- The anticipated mean difference in the standard deviation between the two groups will be 3.2¹⁰

In order to detect this difference of 1.4 points in the mean SOFA score (at a significance level of 5% with 90% power using a two-sided t-test for independent groups, 117 patients will be needed per trial arm (software: nQuery Advisor 6.02 Statistical Solutions). Considering a drop-out rate of about 15%, a total of **280 patients need to be randomized**.

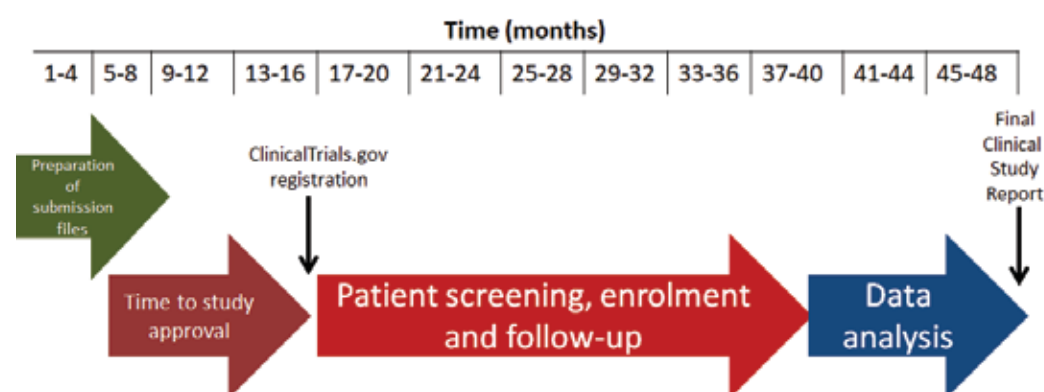


Figure 1: The flow of the ImmunoSep study

STATISTICAL ANALYSIS

The statistical tests per each study endpoint will be the following:

- Primary endpoint decrease of SOFA score: Welch's t-test for mean differences
- Secondary endpoint 28-day mortality: Fisher exact test and survival log-rank test; odds ratio and 95% confidence intervals will be calculated by Mantel-Haenszel's test
- Secondary endpoint 90-day mortality: Fisher exact test and survival log-rank test; odds ratio and 95% confidence intervals will be calculated by Mantel-Haenszel's test
- Secondary endpoint Impact of personalized immunotherapy on the reversal of hyper-inflammation or immunoparalysis: Fisher exact test; odds ratio and 95% confidence intervals will be calculated by Mantel-Haenszel's test
- Exploratory endpoint Impact of personalized immunotherapy on infection resolution: Fisher exact test; odds ratio and 95% confidence intervals will be calculated by Mantel-Haenszel's test
- Exploratory endpoint development of surrogate genomic and proteomic biomarkers: bioanalysis from partners of the ImmunoSep project

ADVERSE EVENTS

Adverse event reporting

Treatment-emergent adverse events (TEAEs) and Serious TEAEs (STEAEs) will be captured from baseline until the last patient's evaluation. Investigators should monitor subjects for TEAEs and are responsible for recording ALL adverse events and adverse reactions occurring to a patient during the trial. A TEAE is any undesirable and unintended medical occurrence in a subject administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment. The TEAEs may be a sign, a symptom or an abnormal laboratory finding.

The time relationship is defined from the moment the TEAE occurs during therapeutic treatment until 30 days or 5 half-lives after treatment discontinuation. After that any adverse reaction is recorded as adverse event/serious adverse event (AE/SAE) respectively.

If a TEAE/AE meets any of the following criteria, it is considered serious (STEAE/SAE):

- **Life-threatening situation.** The subject was at risk of death at the time of the adverse event/experience. It does not refer to the hypothetical risk of death if the adverse event/adverse reaction were more severe or were to progress.
- **In-patient hospitalization or prolongation of existing hospitalization.**
- **Persistent or significant disability/incapacity.** Any TEAE/AE having an outcome that is associated with a substantial disruption of the ability to carry out normal life functions, including the ability to work. This is not intended to include transient interruption of daily activities.
- **Congenital anomaly/birth defects.** Any structural abnormality in subject's offspring that occurs after intrauterine exposure to treatment.
- **Important medical events/experiences** that may not result in death, be life-threatening, or require hospitalization may be considered a serious adverse event/experience when, based upon appropriate medical judgment, **they may jeopardize the subject and may require medical or surgical intervention to prevent one of the outcomes listed above**, i.e., death, a life-threatening adverse event/experience, inpatient hospitalization or prolongation of existing hospitalization, a persistent or significant disability/incapacity, or a congenital anomaly/birth defect. Examples of such medical events/experiences include allergic bronchospasm requiring intensive treatment in an emergency room or at home, blood dyscrasias or convulsions that do not result in inpatient hospitalization, or the development of drug dependency or drug abuse.
- **Spontaneous and elective abortions** experienced by study subject.

Grading of severity

The severity of the TEAE/AE shall be graded as:

- **Mild:** the TEAE/AE is transient and well tolerated by the patient
- **Moderate:** the TEAE/AE causes discomfort and affects the usual activities of the patient.
- **Severe:** the TEAE/AE affects the usual activities of the patient to an important degree and may cause disability or be life-threatening.

Relationship to the drug

The investigator will use the following definitions to assess the relationship of the adverse event to study drug:

- **Probably Related:** The TEAE/AE has a strong time relationship to the drug or relapses if re-induced, and another etiology is improbable or clearly less probable.
- **Possibly Related:** The TEAE/AE has a strong time relationship to the drug and an alternative aetiology is as probable or less probable.
- **Probably not Related:** The TEAE/AE has a slight or no time relationship to the drug and/or there is a more probable alternative aetiology.
- **Unrelated:** The TEAE/AE is due to an underlying or concomitant disease or to another pharmaceutical product and is not related to the drug (no time relationship and a much more probable alternative aetiology).

If an investigator's opinion of possibly related, probably not related or not related to study drug is given, an alternate etiology must be provided by the investigator. Please note that a severe TEAE/AE is not necessarily serious, as the term severe is a measure of intensity. Individual un-blinding thought to be necessary for the management of an adverse event will be documented in the subject Case Report Form. All Investigators must report every TEAE/AE and evaluate the severity and possible causality with the study drug according to aforementioned criteria. All TEAEs/AEs are reported to Sponsor. The sponsor is responsible for evaluation of all TEAEs/AEs. All STEAEs/SAEs must be reported within 24 hours by completion of the TEAE/AE and faxed to the Hellenic Institute for the Study of Sepsis. The Sponsor must evaluate whether a TEAE is expected or not. A STEAE/SAE may qualify for expedited reporting to regulatory authorities if it is determined to be a suspected, unexpected serious adverse reaction (SUSAR). The Sponsor is responsible for submitting expedited safety reports to the appropriate regulatory agency for all confirmed SUSARs. In the case of a fatal or life-threatening SUSAR, the Sponsor will notify the appropriate regulatory agency as soon as possible but in no case later than 7 calendar days after the Sponsor's initial receipt of the information. For a non-life-threatening SUSAR, the report will be submitted no later than 15 days after the Sponsor is made aware of the event. The Sponsor has the obligation to submit

annually a drug safety updated report (DSUR) according to global experience to appropriate regulatory authorities. The electronic submission to Eudravigilance will be performed through the organization ID: HISS.

QUALITY CONTROL AND ASSURANCE

Quality control and assurance checks are performed by sponsor in order to allow periodic review of adequacy of the study activities and practices and allow for revising for revising such practices as needed so the data and process are maintained, the study meets the protocol and procedural requirements, and is reproducible.

Before enrolling any subjects in this study, sponsor personnel and the investigator review the protocol, the IB, the CRFs and instructions for their completion, the procedure for obtaining informed consent, and the procedure for reporting AEs and SAEs.

A qualified representative of the sponsor monitors the conduct of the study by visiting the site and by contacting the site by telephone and e-mail. During these site visits, all source documents are reviewed and information recorded in the CRFs is verified against them.

All collected data coming from and study visits will be uploaded to the electronic Case Report Form (eCRF) system of HISS. The platform is safe and it is based on the availability of a personal server (www.sepsis.gr). Each principal investigator and each sub-investigator who remain blinded to the allocated study drug will be provided a unique username and password allowing uploading of data to the study eCRF platform. The same platform will be used by the CRAs for the monitoring.

Study monitoring plan / Inspection

Clinical research associates (CRAs) of the Sponsor will establish and maintain regular contact between the investigator and the sponsor. CRAs will evaluate the competence of each study center, informing the sponsor about any problems relating to facilities, technical equipment or medical staff. During the study, CRAs will check that written informed consent has been obtained from all subjects correctly according to principles of Declaration to Helsinki, GCP, and applicable regulatory requirements and that data are recorded correctly and completely. CRAs are also entitled to

compare entries in CRFs with corresponding source data and to inform the investigator of any errors or omissions. CRAs will also control adherence to the protocol at the investigator site. They will arrange for the supply of investigational product and ensure appropriate storage conditions are maintained. The CRAs will make written reports to the sponsor on each occasion when contact with the investigator is made, regardless of whether it is by e-mail or in person. During monitoring visits, source data verification will be carried out by CRAs and all entries in the CRFs will be compared with the original source documents. The investigators must agree to meet with the CRAs at regular intervals and to cooperate in resolving any queries or findings made during the monitoring process.

Beside routine monitoring quality assurance will be documented through independent auditing of the quality control activities and where applicable, by regulatory authorities through inspections. Subject confidentiality will be protected at all times.

ETHICAL CONSIDERATIONS

Prior to the initiation of this study, the study design will receive ethical, scientific, and where applicable, regulatory review. Investigators will conduct this study in accordance with the principles of the Declaration of Helsinki, GCP, and applicable regulatory requirements.

Regarding Informed Consent Form obtaining procedures, any procedures before any procedures specified in the protocol are performed, a subject must:

- Be informed of all pertinent aspects of the study and all elements of informed consent
- Be given time to ask questions and time to consider the decision to participate
- Voluntarily agree to participate in the study
- Sign and date the updated and approved by IRB/IEC/REB ICF version.

PROTOCOL ADHERENCE AND AMENDMENTS

Investigators ascertain that they will apply due diligence to avoid protocol deviations. All significant protocol deviations will be recorded and reported in the clinical study report (CSR). Any change or addition to the protocol can only be made

in a written protocol amendment that must be approved and signed by the sponsor,
health authorities where required, and the IRB/IEC/REB.

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APPENDIX I Final study report of the PROVIDE study

**A PERSONALIZED RANDOMIZED TRIAL OF VALIDATION AND RESTORATION
OF IMMUNE DYSFUNCTION IN SEVERE INFECTIONS AND SEPSIS (THE
PROVIDE TRIAL)**

FINAL STUDY REPORT

Evangelos J. Giamarellos-Bourboulis, MD, PhD

Mihai G. Netea, MD, PhD

Authors

Sponsor and CRO:

Hellenic Institute for the Study of Sepsis

88 Michalakopoulou str., 11528 Athens, Greece

insepsis@otenet.gr

tel 00302107480662

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INTRODUCTION

The definition of sepsis as a life-threatening organ-function dysfunction due to the dysregulated response of the host upon an infection introduced in 2016¹ was based on lot of research efforts conducted the last three decades targeting to understand the dysregulation of the response of the host. Several large-scale randomized clinical trials with immunomodulators have failed over the last two decades² leading to understand that a new approach is needed². More precisely, we know nowadays that patients can have completely different mechanisms mediating organ dysfunction although they present with similar clinical findings. As a consequence, we need to understand early which patient is suffering from which type of immune dysregulation and intervene to modulate this precise dysregulation and abandon the traditional approach of all patients receiving the same adjunctive treatment. One such example is that 3 to 4% of patients have hyper-inflammation associated with high risk for death the very first 10 days. These patients have excess production of interleukin (IL)-1 β by tissue macrophages, leading to pancytopenia, bone marrow haemophagocytosis, liver dysfunction and disseminated intravascular coagulation (DIC) conditions resembling with macrophage activation-like syndrome (MALS)³. High values of ferritin are associated with high specificity for the diagnosis of MALS⁴, and anakinra, one recombinant antagonist of the IL-1 receptor, may be a treatment of choice⁵.

Another example are patients who present with “immunological exhaustion” or immunoparalysis⁶. The cells of the immune system become dysfunctional, resulting to susceptibility for secondary infections, prolonged hospitalization and increased mortality. It is suggested that the decrease of the expression of the human leukocyte antigen (HLA)-DR expression on the membrane of circulating monocytes is the hallmark of immunoparalysis and that this can be reversed by recombinant human interferon- γ (rhIFN γ)⁷.

PROVIDE (a Personalized Randomized trial Of Validation and restoration of Immune Dysfunction in severE infections and Sepsis) was a randomized clinical trial that was conducted with two main objectives: the first was to validate that the suggested immune classification into MALS, immunosuppression and unclassified is correct and reflects final outcome. The second was to anakinra or rhIFN γ in patients with MALS and immunoparalysis respectively and to investigate the impact on the outcome.

PATIENTS AND METHODS

PROVIDE was a double-blind, double-dummy randomized clinical trial that took place between November 2017 and December 2019 in 14 study sites in Greece (nine Intensive Care Units and five departments of Internal Medicine) (EudraCT 2017-002171-26; approval by the National Ethics Committee of Greece 78/17; approval IS 75-17 by the National Organization for Medicines of Greece; Clinicaltrials.gov registration NCT03332225). Participants were adults (age ≥ 18 years) of both genders with written Informed Consent Form (ICF) provided by themselves or legal representatives with lower respiratory tract infection (community-acquired pneumonia or hospital-acquired pneumonia or ventilator-associated pneumonia) or primary bacteremia or acute cholangitis, and sepsis. Infections were defined according to international criteria (see Supplement)⁸. Sepsis was defined as total score of 2 or more points of the Sequential Organ Failure Assessment (SOFA) score for new admissions or as 2 or more points increase of the total SOFA score for hospitalized patients. Septic shock was defined as persisting mean arterial pressure below 65mmHg despite adequate intravenous administration fluids with plasma lactate greater than 2 mmol/l necessitating treatment with vasopressors. Exclusions criteria were: other type of infections, stage IV malignancy, HIV infection, primary immunodeficiency, chronic intake of corticosteroids defined as total daily dose equal or greater than 0.4mg/kg of equivalent prednisone for more than the last 15 days, intake of anti-cytokine biological agents, history of systemic lupus erythematosus, history of demyelinating disorders, pregnancy, lactation and decision of no resuscitation.

PROVIDE has one *screening stage* and one *intervention stage*. During the *observation stage* there was a daily recording of the following parameters for 28 consequent days: type of infection, co-morbidities, results of blood cell count, coagulation tests, renal and hepatic biochemistry, arterial blood gases, culture results of blood, urine and tracheo-bronchial secretions, imaging results and overall survival. Each patient had a blood draw of 10ml via a peripheral vein within the first 24 hours from study inclusion (day 1) which was repeated after 24 hours (day 2). Samples were collected in EDTA-containing and clean vials and transferred via courier to the central lab. Ferritin was measured in serum by an enzyme immunoassay (ORGENTEC Diagnostika GmbH, Mainz, Germany; lower detection limit 75ng/ml).

Using whole blood from the EDTA via,; white blood cells were incubated for 15 minutes in the dark with the following monoclonal antibodies: anti-CD14 FITC, anti-HLA-DR-PE, anti-CD45 PC5 (Immunotech, Marseille, France) and Quantibrite HLA-DR/anti-monocyte PerCP-Cy5 (Becton Dickinson, Cockeysville Md). After incubation, cells were analysed by flow cytometer FC500 versus cells stained with anti-CD45 PC5 and anti-idiotypic IgG1. HLA-DR molecules on CD14/CD45 monocytes were expressed as a percentage of positive stained cells and as number of molecules per cell. MALS was defined as ferritin more than 4.420ng/ml on both samples or on day 2 sample regardless of flow cytometry results. Immunoparalysis was diagnosed as expression of HLA-DR/CD14/CD45 less than 30% on of both days on the sample of day 2, provided that ferritin concentration was $\leq 4.420\text{ng/ml}$. Patients that could not be classified as MALS or immunoparalysis were considered having an intermediate immune status. Additionally, peripheral blood mononuclear cells (PBMCs) were isolated from each EDTA blood sample using gradient centrifugation over Ficoll-Hypaque density gradient. After three washings in ice-cold PBS (phosphate buffered saline, pH: 7.2), PBMCs were counted using a Neubauer plate with trypan blue exclusion of dead cells and distributed on 96-wells plate at final volume 200 μl and density $2.5 \times 10^6/\text{ml}$. PBMCs were incubated in RPMI 1640 (Biochrom, Berlin, Germany) enriched with 2 mM of L-glutamine, 100 U/ml penicillin G, 0.1 mg/ml streptomycin without/with 10 ng/ml lipopolysaccharide (LPS) of *Escherichia coli* O55:B5 and 5 $\mu\text{g/ml}$ phytohaemagglutinin (PHA). After incubation for 48 hours at 37°C and 5% CO₂ the supernatants were collected and tumor necrosis factor- α (TNF- α) concentration was measured in duplicate using an enzyme immunosorbent assay (Bender MedSystems GmbH, Vienna, Austria; lower detection limit 20 pg/ml).

Patients diagnosed with septic shock and MAS or immunoparalysis were allowed to participate in the *intervention stage* of the randomized clinical trial. They were randomized into adjunctive treatment either with personalized immunotherapy or placebo. Patients randomized to personalized immunotherapy were administered anakinra (Kineret, Swedish Orphan Biovitrum, Athens) 200 mg intravenously (iv) at 20ml every 8 hours (q8h) for 7 days and 0.5ml of NaCl 0.9% subcutaneously (sc) every 48 hours (q48h) for 15 days, if classified with MALS; if classified with immunoparalysis they were administered 20ml of NaCl 0.9% iv q8h for 7 days and 100 μg rhIFN γ (Immukin, Boehringer Ingelheim, Athens) at 0.5 ml sc q48h for 15 days. Patients randomized to placebo arm were administered NaCl 0.9% iv and sc as

described above. The above dose of anakinra was adjusted to 100 mg q8h when creatinine clearance was less than 30 ml/min. Adverse events were daily captured (see supplement).

The endpoint of the *observation stage* was the association of the studied immunological classification with 28-day outcome. The primary study endpoint of the *intervention stage* was the effect of personalized immunotherapy in 28-day mortality. Secondary endpoints were 90-day mortality, secondary infections and the appropriateness of immune classification using the total of screened patients from the original screening stage. Exploratory endpoint was the change of SOFA score among survivors at the end of treatment. Using the hypothesis 28-day mortality of immunotherapy would decrease mortality from MALS from 60% to 35% and mortality from immunoparalysis from 50% to 35%, it was calculated that 278 patients should be enrolled to demonstrate these differences with 80% power at the 10% level of significance.

Qualitative variables were expressed as percentages and 95% confidence intervals (CIs) and compared by the Fisher's exact test. Quantitative variables were expressed as mean $\pm$ SD (standard deviation) or SE (standard error) and compared was done using the Student t-test or the Mann Whitney U. The prognostic output of the percentage and of absolute counts of HLA-DR molecules on monocytes for 28-day mortality was studied by ROC (receiver operator characteristics) curve analysis. The co-ordinates showing the highest sensitivity and specificity (Youden index) were selected. Multivariate step-wise Cox regression analysis was used to validate if immunological classification was an independent factor of unfavorable outcome; relative risk (RR) and 95% confidence intervals (CIs) were calculated. Any p value lower than 0.05 was considered statistically significant.

RESULTS

The flow chart of both stages of PROVIDE is presented in Figure 1. During the first stage, it was found that the frequency of MALS was higher among patients with septic shock (44 out of 177 patients, 24.9%) compared to patients without septic shock (4 out of 63 patients, 6.3%, p : 0.001 compared to septic shock). When the entire cohort of 240 studied patients was split into those with MALS (as diagnosed with ferritin >4420 ng/ml) and into those without MALS (as diagnosed with ferritin ≤ 4420 ng/ml), it was found that the characteristics of liver dysfunction and

disseminated intravascular coagulation were more common in MALS. More precisely, patients with MALS had lower absolute counts of platelets, higher international normalized ratio (INR), prolonged activated partial thromboplastin time (APTT), and higher levels of aspartate aminotransferase (AST), aspartate aminotransferase (AST) and total bilirubin. These findings confirm that ferritin >4.420 ng/ml are consistent with the characteristics of macrophage activation syndrome.

Surprisingly, only two patients were diagnosed with immunoparalysis. This led us to consider that the percentage of the HLA-DR/CD14/CD45 is not the most appropriate tool for diagnosis. Indeed, after ROC curve analysis it was found that the prognostic performance of the percentage of HLA-DR/CD14/CD45 for 28-day mortality was poor (Figure 2A). The same analysis was repeated using the number of HLA-DR receptors on CD14/CD45 monocytes as predictor and the prognostic performance was substantially improved (Figure 2B). Using Youden index analysis, a cut-off of HLA-DR receptors less than 5000/monocyte was found to be associated with better prediction of death after 28 days so as to be considered as diagnostic tool of immunoparalysis (Figure 2C). This cut-off was confirmed in two approaches: the first was to find that patients with less than 5000 HLA-DR/monocyte had lower production capacity for TNF α by their PBMCs (Figure 2D). The second approach was by finding that the laboratory findings of impaired coagulation and liver dysfunction were less evident among patients with less than 5000 HLA-DR/monocyte than among patients with MALS (Figure 2E).

Among patients diagnosed with neither MAS nor immunoparalysis, 21 (23.6%) had ferritin between 1000 and 4420 ng/ml that is consistent with intense inflammatory reaction¹⁰. Also 23 patients (25.9%) had HLA-DR receptors between 5000 and 8000/monocyte that is consistent with partial dysfunction of the immune system¹¹. Therefore, these patients were classified to an intermediate immune status.

As a consequence, the enrolled 240 patients were classified into three groups of immune activation: MALS group (n=48) (20.0%), immunoparalysis (n=103) (42.9%) and intermediate immune state (n=89) (37.1%). The repeatability of this classification of the first and second day of sampling was 88.5%. Comparison of the clinical characteristics of these three groups is presented in Table 1. Apart from larger values on severity scores and greater frequency of septic shock in MALS, no other differences were found. The classification groups differed significantly on 28-day mortality. Mortality of MALS group was 79.1% (95% CIs 65.7-88.3%); of

immunoparalysis 66.9% (95%CI 56.4-74.4%); and of intermediate state 41.6% (95%CI 31.9-51.9%) (Figure 3A). Multivariate analysis showed that this immune classification was an independent predictor of 28-day mortality ($HR_{\text{immunoparalysis VS intermediate}} 1.79$; 95% CI 1.19-2.68; and $HR_{\text{MALS VS intermediate}} 2.39$; 95% CI 1.49-3.84) regardless of severity scores and presence of septic shock (Figure 3B). It is important to state that the antimicrobials administered were similar in all classification groups, therefore difference on survival is not attributed to medical treatment.

As presented on Figure 1, 36 patients were enrolled to the double-blind, double-dummy randomized clinical stage of PROVIDE and were randomized to administration of personalized immunotherapy (n=15) or of placebo (n=21) (Table 2). The first patient was randomized on 2 December 2017 and the last on 1 December 2019. By December 2019 it was decided to prematurely stop the study due to the slow enrolment of patients. The enrolment rate was very slow which was due to the fact that 30% cut-off of HLA-DR/CD14/CD45 that was used as a tool of enrolment in the study for patients with immunoparalysis was found in only two patients. One patient per treatment arm was diagnosed with immunoparalysis; all the remaining enrolled patients were diagnosed with MALS.

28-day survival did not differ (Figure 4A); 18 deaths occurred in the placebo arm (28-day mortality 85.7%; 95%CI 65.4-95.1%); and 14 deaths in the personalized immunotherapy arm (28-day mortality 93.3%; 95%CI 70.2-98.8%). Mortality after 90 days was the same. No difference was found between the two groups on the incidence of secondary infections (see Table 3 for adverse events).

The study exploratory endpoint was survival state associated with SOFA change at the end of treatment. Since only one patient per group had immunoparalysis, so that the actual studied treatment was anakinra, this endpoint was analysed by day 7 when anakinra treatment ended. Indeed, by day 7 five patients allocated to placebo were alive with no increase of baseline SOFA (25%) compared to 64.3% of anakinra-treated patients (Figure 4B). On the same day 7, there was a significant reduction of ALT and of INR which is the hallmark of MALS (Figure 4C and 4D) pointing towards a positive treatment effect of anakinra of MALS in these patients.

There was no difference in the incidence of adverse events between the two arms of the study (Table 3).

DISCUSSION

This trial is globally the first attempt to study the efficacy of an algorithm of personalized treatment as for the outcome sepsis. This approach is far different from the paradigm existing so far which is based on the post-hoc identification of subgroup of patients receiving most of benefit. Evaluation of personalized treatment mandates the direct comparison of two groups; one group receiving the standard-of-care and placebo adjunctive treatment; and another receiving different types of targeted immunotherapy guided by biomarkers group. PROVIDE is the first trial that is managing to do so. The main success of PROVIDE is the development of a system of classification of three strata of immune activation in sepsis using ferritin and the number of HLA-DR receptors on monocytes. Classified states of MALS, immunoparalysis and intermediate are characterized by distinct risk of death.

MALS resembles secondary hemophagocytic lymphohistocytosis following severe infections dominated by hyper-production of pro-inflammatory cytokines¹². A retrospective analysis of a phase III clinical trial of the efficacy of intravenous anakinra in severe sepsis¹³ failed to show any survival benefit. However, post-hoc analysis showed 30% reduction of 28-day mortality in the small subgroup of patients with liver dysfunction and (DIC) who most likely had MALS⁵. In this study anakinra was administered daily for 28 days contrary to PROVIDE where the total duration of anakinra treatment was limited to seven days.

The importance of immunoparalysis was revealed by the study of monocytes and lymphocytes isolated from the spleen of fresh cadavers¹⁴. Yet this study left unanswered if patients died because of immunoparalysis of late onset or if immunoparalysis was a dominant characteristic occurring from the clinical onset. The measurement of absolute number of HLA-DR molecules on monocytes showed that immunoparalysis is happening in 43% of patients from sepsis onset and that it is evenly appearing in sepsis without shock and in septic shock. Immunostimulation is an attractive treatment strategy for immunoparalysis¹⁶. In a small open-label non-randomized clinical trial, rhIFN γ treatment was administered in nine patients with septic shock. Reversal of immunoparalysis was achieved to all nine with increase of expression of HLA-DR molecules on monocytes alongside ex vitro increase of TNF α production following stimulation by LPS; clinical resolution occurred in eight patients¹⁷. rhIFN γ was administered in six healthy volunteers with immunoparalysis

induced in a model of experimental endotoxemia. Volunteers receiving rhIFN γ sc experienced significant increase of HLA-DR expression on monocytes⁷.

Due to the diagnostic limitation of percent expression of HLA-DR/CD14/CD15 cells to recognize immunoparalysis, few patients were randomized to rhIFN γ treatment or matched placebo. As such, PROVIDE practically reported the therapeutic efficacy of seven days of intravenous treatment with anakinra. This intervention altered the clinical course of the first seven days by reducing mortality and increase of SOFA score. In parallel laboratory signs of MALS were improved. The lack of any difference in terms of 28-day survival might be attributed to the higher rate of chronic renal failure as a co-morbidity among patients randomized to the immunotherapy arm.

PROVIDE managed to define three distinct function levels of immune classification of sepsis. Two of these three levels, namely MALS and immunoparalysis, may be part of an algorithm of personalized immunotherapy. Anakinra treatment was promising to improve the clinical course of sepsis.

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Table 1 Patients characteristics according to immune classification

	Intermediate stats (N=89)	Immunoparalysis (N=103)	MALS (N=48)	p
Age, years, mean $\pm$ SD	71.7 $\pm$ 16.5	74.2 $\pm$ 12.5	70.8 $\pm$ 14.2	0.308*
Male gender, n (%)	50 (56.2)	63 (61.2)	29 (60.4)	0.767**
CCI, mean $\pm$ SD	4.8 $\pm$ 2.3	5.2 $\pm$ 2.4	4.9 $\pm$ 2.3	0.398*
Septic shock, n, %	55 (61.8)	78 (75.7)	44 (91.7)	0.001**
APACHE II score, mean $\pm$ SD	21.1 $\pm$ 8.7	24.9 $\pm$ 8.5	29.5 $\pm$ 9.2	<0.0001*
SOFA score, mean $\pm$ SD	9.4 $\pm$ 4.0	11.0 $\pm$ 3.8	13.8 $\pm$ 3.5	<0.0001*
CRP, mg/l, mean $\pm$ SD	108.4 $\pm$ 105.7	108.6 $\pm$ 102.1	96.8 $\pm$ 104.8	0.843*
Type of infection, n, %)				
Community acquired pneumonia	40 (44.9)	45 (43.7)	19 (39.6)	0.740**
Hospital acquired pneumonia	21 (23.5)	21 (20.4)	12 (25.0)	0.407**
Ventilator associated pneumonia	8 (9.0)	15 (14.6)	5 (10.4)	0.465**
Primary bacteremia	13 (14.6)	16 (15.5)	10 (20.8)	0.620**
Acute cholangitis	7 (7.9)	6 (5.8)	2 (4.2)	0.676**
Gram-negative pathogens (n, %)	19 (21.3)	23 (19.4)	11 (22.9)	0.825**
Pathogens, n (%)				
<i>Escherichia coli</i>	1 (1.1)	1 (0.9)	1 (2.1)	0.840**
<i>Acinetobacter baumannii</i>	3 (3.3)	7 (6.7)	1 (2.1)	0.338**
<i>Klebsiella pneumoniae</i>	5 (5.6)	2 (1.9)	4 (8.4)	0.182**
<i>Staphylococcus aureus</i>	2 (2.2)	6 (5.8)	2 (4.2)	0.465**
<i>Candida albicans</i>	4 (4.5)	2 (1.9)	2 (4.2)	0.578**
Main co-morbidities (n, %)				
Type 2 diabetes mellitus	27 (30.7)	27 (26.2)	16 (33.3)	0.639**
Chronic heart failure	17 (19.1)	28 (27.2)	17 (35.4)	0.105**
Chronic renal disease	9 (10.1)	12 (11.7)	5 (10.4)	0.938**
Chronic obstructive pulmonary disease	21 (23.6)	25 (24.3)	12 (25.0)	0.983**
Administrated antimicrobials (n, %)				
Piperacillin/tazobactam	35 (42.7)	37 (37.8)	13 (27.7)	0.236**
Carbapenems	25 (30.5)	34 (34.3)	21 (44.7)	0.261**
Tigecycline	11 (13.4)	19 (19.4)	12 (25.5)	0.223**

Colistin	25 (30.5)	34 (34.3)	25 (53.2)	0.029**
Vancomycin	17 (20.7)	24 (24.2)	9 (19.1)	0.690**

*comparison using ANOVA test, ** comparison using Pearson chi-square test

Abbreviations APACHE: acute physiology and chronic health evaluation, CCI: Charlson's co-morbidity index, CRP: C- reactive protein, SD: standard deviation, SOFA sequential organ failure assessment

Table 2 Clinical characteristics of treatment groups

	Placebo (n=21)	Personalized immunotherapy (n=15)	p
Age, years, mean $\pm$ SD	69.7 $\pm$ 10.8	70.6 $\pm$ 15.1	0.838*
Male gender, n (%)	12 (57.1)	10 (66.7)	0.732**
CCI, mean $\pm$ SD	4.3 $\pm$ 2.2	5.7 $\pm$ 2.2	0.062*
APACHE II score, mean $\pm$ SD	18.2 $\pm$ 8.7	30.5 $\pm$ 9.4	0.376*
SOFA score, mean $\pm$ SD	14.5 $\pm$ 2.8	14.3 $\pm$ 3.1	0.843*
Type of infection, n (%)			
Community acquired pneumonia	10 (47.6)	2 (13.3)	0.040**
Hospital acquired pneumonia	6 (28.6)	8 (58.3)	0.175**
Mechanical ventilation associated pneumonia	3 (14.3)	3 (20.0)	0.677
Other	2 (13.3)	2 (13.3)	1.00
Main co-morbidities, n (%)			
Type 2 diabetes mellitus	6 (28.6)	6 (40.0)	0.499**
Chronic heart failure	4 (19.0)	8 (53.3)	0.071**
Chronic renal disease	0 (0)	4 (26.6)	0.023**
Chronic obstructive pulmonary disease	7 (33.3)	3 (20.0)	0.468**
Administrated antimicrobials (n, %)			
Piperacillin/tazobactam	3 (14.3)	2 (13.3)	1.00**
Carbapenems	10 (47.6)	12 (80.0)	0.083*
Tigecycline	6 (28.6)	3 (20.0)	0.705**
Colistin	13 (61.9)	10 (66.7)	1.00*
Vancomycin	3 (14.3)	4 (26.7)	0.418**
Antimicrobial treatment according to ESCMID guidelines	18 (85.7)	12 (80.0)	0.677**
Hydrocortisone replacement	19 (90.5)	10 (80.0)	0.639**

* comparison by the Student's t-test, ** comparison by the Fisher exact

Abbreviations APACHE: acute physiology and chronic health evaluation, CCI: Charlson co-morbidity index, ESCMID: European Society of Clinical Microbiology and Infectious Diseases; SD: standard deviation, SOFA sequential organ failure assessment

Table 3 Adverse events captured during the randomized clinical trial

	Placebo (n=21)	Personalized immunotherapy (n=15)	p-value*
Incidence of at least one SAE**, n (%)	4 (19.0)	4 (16.7)	0.694
Ventricular tachycardia	0 (0)	2 (13.3)	0.167
Digital ischemia	0 (0)	2 (13.3)	0.167
Acute myocardial infarction	1 (4.8)	0 (0)	1.00
Brain tumor metastasis	1 (4.8)	0 (0)	1.00
Obstructive ileus	1 (4.8)	0 (0)	1.00
Pulmonary embolism	1 (4.8)	0 (0)	1.00
At least one AE, n (%)	2 (9.5)	0 (0)	0.500
Grade I hyperkalemia	1 (4.8)	0 (0)	1.00
Grade III increase of γ GT	1 (4.8)	0 (0)	1.00

*this does not include the number of deaths

**by the Fisher exact test

Abbreviation AE: adverse event; γ GT: gamma glutmayl transpeptidase; SAE: serious adverse event

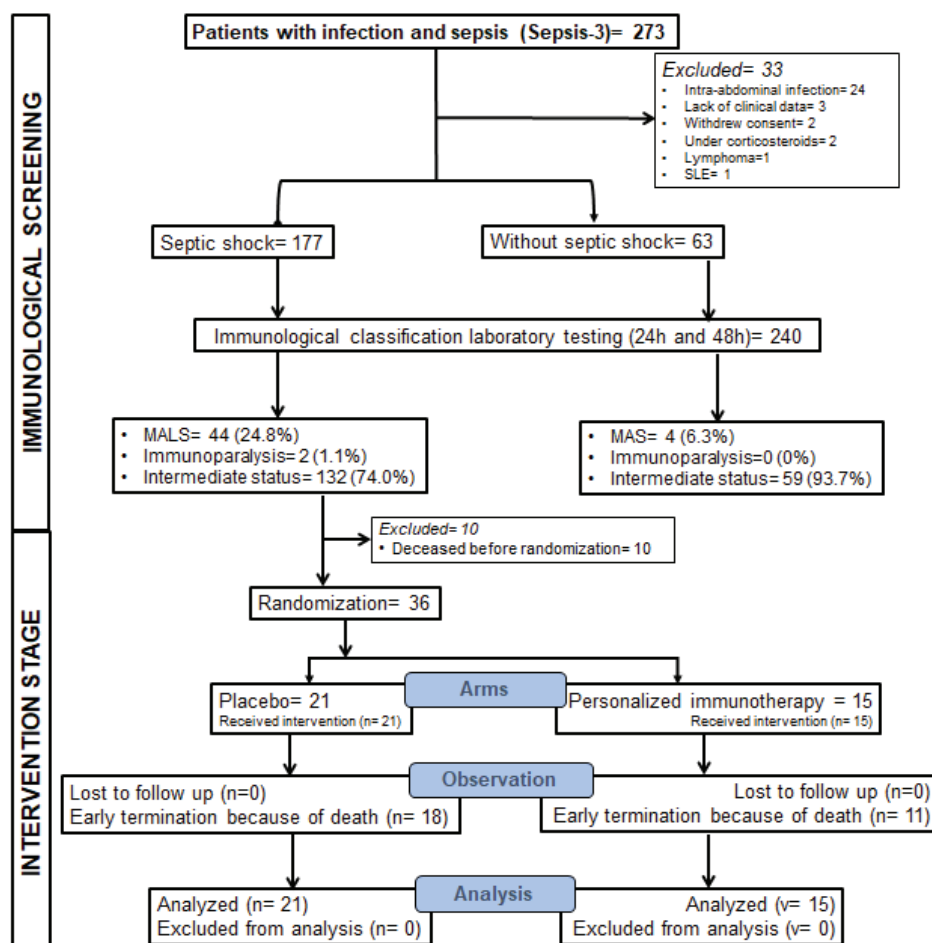


Figure 1 Flow of patients among the two stages of PROVIDE

The first stage was aiming to classify patients with septic shock and without septic shock into MALS (macrophage activation-like syndrome), immunoparalysis and intermediate. The second intervention stage was a double-blind, double-dummy randomized clinical trial where patients with septic shock and MALS or immunoparalysis received blind adjunctive treatment with placebo or personalized immunotherapy

Abbreviation SLE: systemic lupus erythematosus

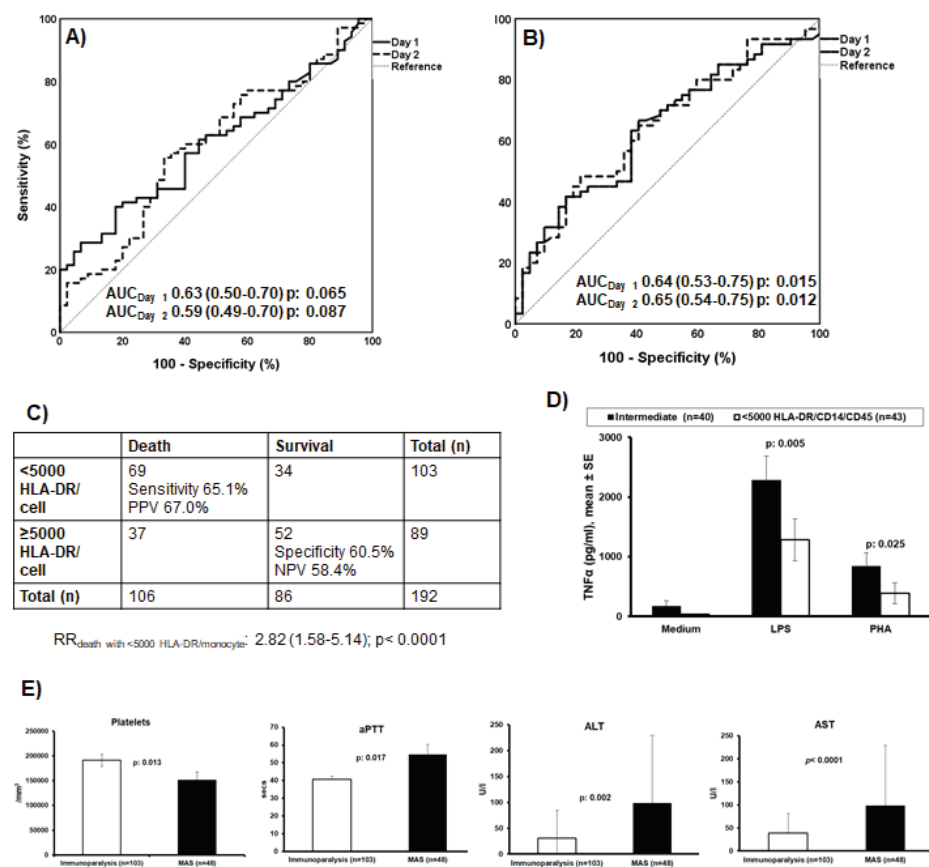


Figure 2 Development of the number of HLA-DR receptors/monocyte as diagnostic tool for sepsis-induced immunoparalysis

ROC curve (A) of percentage and (B) of number of HLA-DR receptors /monocyte of each day (Day 1 and Day 2) as a mortality predictor.

(C) Prognostic performance of <5000 HLA-DR molecules /monocyte for 28-day mortality. (D) Comparison the capacity ability of peripheral blood mononuclear cells for tumor necrosis factor-alpha (TNF-α) production, following stimulation by endotoxin (LPS) and phytohaemagglutinin (PHA), between patients with <5000 HLA-DR molecules /monocyte and with an intermediate status (ferritin ≤4420 ng/ml and ≥ 5000 HLA-DR molecules /monocyte).

(E) Comparison of coagulation parameters and liver function enzymes levels between immunoparalysis group and macrophage activation syndrome (MAS) group. The p values of the indicated comparisons are shown.

Abbreviations ALT: alanine aminotransferase, AST: aspartic aminotransferase, AUC: area under curve, aPTT: activated partial thromboplastin time, NPV: negative predictive value, CI: confidence interval, PPV: positive predictive value, RR: relative risk

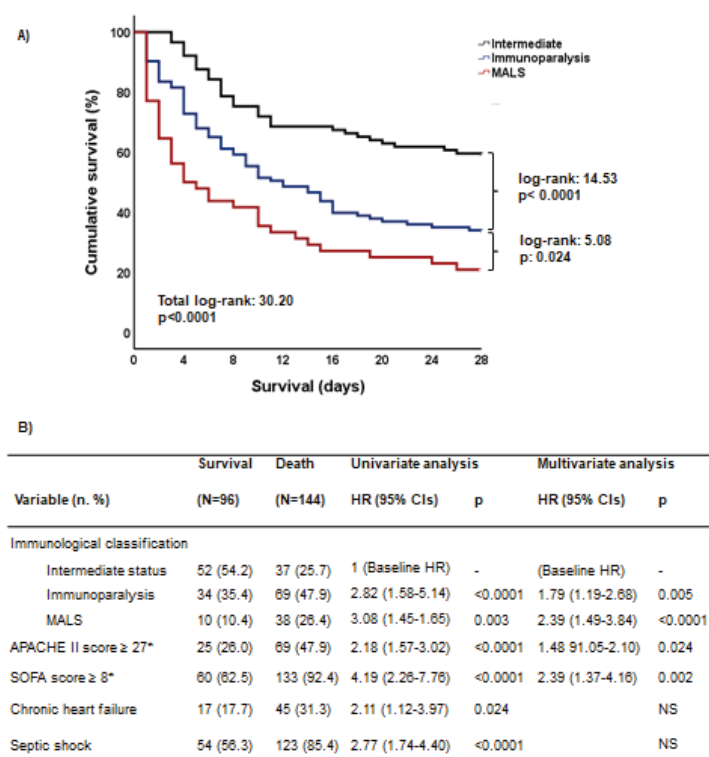


Figure 3 Association of immune classification of sepsis and 28-day mortality

A) Survival curves of each classification group: macrophage activation syndrome (MALS), immunoparalysis and intermediate stats. Log-rank values and p-values are presented. B) 28-day mortality related risk factors after univariate and multivariate Cox regression analysis. Hazard ratios (HR), 95% confidence intervals (CIs) and p-values are presented.

Abbreviations APACHE acute physiology and chronic health evaluation, SOFA sequential organ failure assessment, NS: statistically non significant

*Values represent Youden Index of ROC curves for mortality

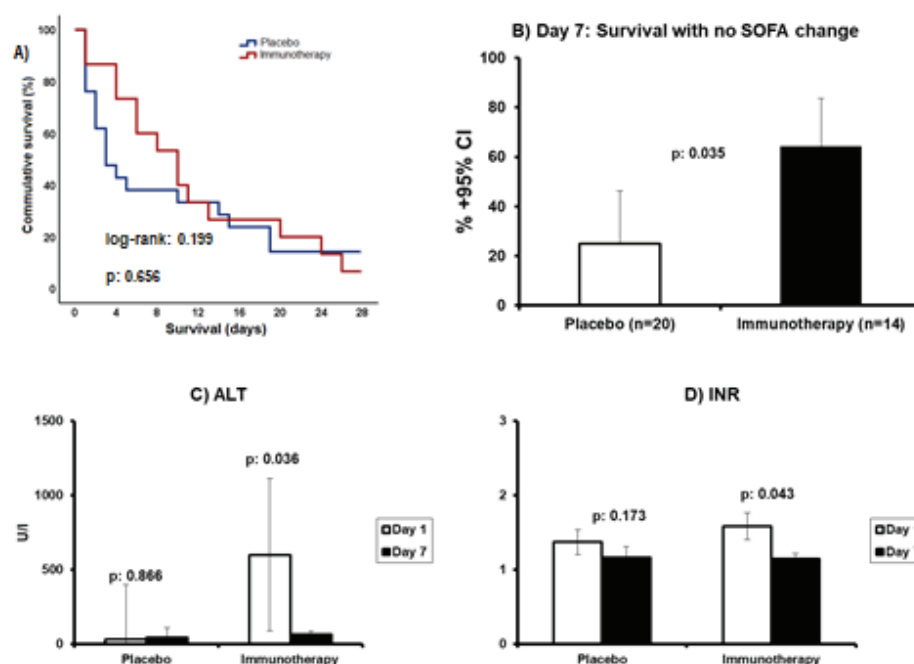


Figure 4 Effect of personalized immunotherapy

A) Survival curves of both treatment arms. Log-rank value and p value are presented.

B) Comparison of the rate of survivors without increase of SOFA score by day 7.

C and D) Comparison of alanine aminotransferase (ALT) and of international normalized ratio (INR, graph D) values between days 1 and 7 for each treatment arms; p values of indicated statistical comparison are presented.

Abbreviation CI: confidence interval; SOFA: sequential organ failure assessment

APPENDIX II

The list with the study sites that will participate in the ImmunoSep study is as follows:

France

- Intensive Care Unit, Lyon University Hospital (Principal Investigator: Prof. Thomas Rimmelé, MD, PhD)

Germany

- Intensive Care Unit, Jena University Hospital (Principal Investigator: Prof. Michael Bauer, MD, PhD)

Greece

- 4th Department of Internal Medicine, ATTIKON University Hospital, Athens (Principal Investigator: Ass. Prof. Antonios Papadopoulos, MD, PhD)
- 2nd Department of Propaedeutic Internal Medicine, ATTIKON University Hospital, Athens (Principal Investigator: Prof. E. Hatziaggelaki, MD, PhD)
- 2nd Department of Critical Care Medicine, ATTIKON University Hospital, Athens (Principal Investigator: Prof. Apostolos Armaganidis, MD, PhD)
- 3rd Department of Internal Medicine, SOTIRIA General Hospital of Chest Diseases of Athens, Athens (Principal Investigator: Ass. Prof. Garyphallia Poulakou, MD, PhD)
- 1st Department of Pulmonary Medicine and Intensive Care Unit, SOTIRIA General Hospital of Chest Diseases of Athens, Athens (Principal Investigator: Prof. Antonia Koutsoukou, MD, PhD)
- Department of Internal Medicine, Larissa University General Hospital (Principal Investigator: Prof. George Dalekos, MD, PhD)
- Intensive Care Unit, “KAT” General Hospital of Attika, Athens (Principal Investigator: Ioannis Alamanos, MD)
- Intensive Care Unit, Ioannina University General Hospital (Principal Investigator: Prof. Vasileios Koulouras, MD, PhD)
- Intensive Care Unit, Alexandroupolis University General Hospital (Principal Investigator: Director of NHS Vassiliki Theodorou, MD)

- Intensive Care Unit, “G.Gennimatas” Thessaloniki General Hospital (Principal Investigator: Eleni Antoniadou, MD)
- Intensive Care Unit, “Aghios Dimitrios” Thessaloniki General Hospital (Principal Investigator: Glykeria Vlachogianni, MD)
- Intensive Care Unit, “AHEPA” Thessaloniki University General Hospital (Principal Investigator: Eleni Gkeka, MD)
- Intensive Care Unit, “Theageneio” Thessaloniki Anticancer Hospital (Principal Investigator: Souzana Anisoglou, MD)
- Intensive Care Unit, “Ippokrateion” Thessaloniki General Hospital (Principal Investigator: Eleni Mouloudi, MD)
- Intensive Care Unit, “Tzaneio” Piraeus General Hospital (Principal Investigator: Athanasios Prekates, MD)
- Intensive Care Unit, “Koutlimbaneio & Triantafylleio” Larissa General Hospital (Principal Investigator: Apostolos Komnos, MD, PhD)
- Latsio Intensive Care Unit, “Thriasio” Elefsis General Hospital (Principal Investigator: Nikolaos Markou, MD)
- 5th Department of Internal Medicine, “Evangelismos” General Hospital of Athens (Principal Investigator: Dr. Theano Kontopoulou, MD, PhD)

The Netherlands

- Department of Internal Medicine and Infectious Diseases, Amsterdam Medical Center (Principal Investigator: Prof. Joost Wiersinga, MD, PhD)
- Intensive Care Unit, University Medical Center Radboud, Nijmegen (Principal Investigator: Prof. Peter Pickkers, MD, PhD)

Romania

- Department of Internal Medicine and Infectious Diseases, Cluj-Napoca (Principal Investigator: Ass. Prof. Mihaela Lupse, MD, PhD)

Switzerland

- Department of Internal Medicine and Infectious Diseases, Centre Hospitaliaire Universitaire Vaudois, Lausanne (Principal Investigator: Prof. Thierry Calandra, MD, PhD)

Monitor Dr. Antigoni Kotsaki, MD, PhD

e-mail: scra@sepsis.gr tel: +30 6946 637 164; +30 210 5832562

Pharmacovigilance supervisor Ms Areti Voulomenou, MEng, MSc

e-mail: voulomenou@suschem.gr, tel.: +30 210 8252510

APPENDIX III

Definitions of eligible infections Any patient can be classified as suffering from one of the following infections if he/she meets listed criteria 1 to 3.

	Criterion 1	Criterion 2	Criterion 3	Resolution criteria (to be considered for the exploratory study endpoint)
Infection	All the following	At least 2 of the following	At least 1 of the following	
Primary bacteremia ¹¹	- At least one positive blood culture - Failure to identify a primary infection site despite thorough clinical and radiology investigation	None	None	Sterile blood cultures after seven days
Community-acquired pneumonia ¹²	New infiltrate on chest X-ray	- New onset or worsening of cough - Dyspnea - Auscultatory findings compatible with pulmonary consolidation	- Procalcitonin ≥ 0.25 ng/ml - Hypoxemia with partial oxygen pressure ≤ 60 mmHg or oxygen saturation $\leq 90\%$ in ambient air - Respiratory rate ≥ 20 breaths/min	- None of criterion 2 findings making the patient eligible for the study AND - None of criterion 3 findings making the patient eligible for the study
Hospital-acquired pneumonia ¹³	- Onset >48 hours from hospital admission - New or evolving infiltrate on chest X-ray	- New onset or worsening of cough or dyspnea - Purulent tracheobronchial secretions - Auscultatory findings	- Procalcitonin ≥ 0.25 ng/ml - Hypoxemia with partial oxygen pressure ≤ 60 mmHg or oxygen saturation $\leq 90\%$ in ambient air - Respiratory rate ≥ 20 breaths/min	- None of criterion 2 findings making the patient eligible for the study AND - None of criterion 3 findings making the patient eligible for the study

PERSONALIZED IMMUNOTHERAPY IN SEPSIS: A MULTICENTRE AND MULTINATIONAL, DOUBLE-BLIND, DOUBLE-DUMMY RANDOMIZED CLINICAL TRIAL (THE IMMUNOSEP TRIAL)

		compatible with pulmonary consolidation		
Ventilator-associated pneumonia ^{13,14}	- Onset >48 hours from start of mechanical ventilation - New or evolving infiltrate on chest X-ray	- Purulent tracheobronchial secretions - Auscultatory findings compatible with pulmonary consolidation	- Procalcitonin ≥ 0.25 ng/ml - Clinical pulmonary infection score (CPIS)* ≥ 6 (see APPENDIX II below)	- None of criterion 2 findings making the patient eligible for the study - AND - None of criterion 3 findings making the patient eligible for the study

*CPIS: clinical pulmonary infection score

APPENDIX IV The clinical pulmonary infection score (CPIS)¹¹

Variable	0 points	1 point	2 points
Tracheal secretions	Rare	Abundant	Purulent
Leukocyte count (/mm ³)	>4,000 and <11,000	<4,000 and >11,000	<4,000 or >11,000+band forms
Temperature (°C)	>36.5 and <38.4	>38.5 and <38.9	>39 or <36
pO ₂ /FiO ₂ ratio (mmHg)	>240 or ARDS*	-	≤240 and no ARDS
Chest radiograph	No infiltrate	Diffuse infiltrate	Localised infiltrate
Culture of tracheal aspirate	Negative	-	Positive

Each variable is scored between 0 and 2. The CPIS is the sum of the score of each variable

*ARDS: acute respiratory distress syndrome

APPENDIX V The SOFA score¹

Variable	0 points	1 point	2 points	3 points	4 points
PaO ₂ /FiO ₂ (mmHg)	≥400	<400	<300	<200	<100
Platelets (per mm ³)	≥150	<150	<100	<50	<20
Hypotension	MAP≥ 70 mmHg	MAP<70 mmHg	Dobutamine whatever dose	Adrenaline ≤0.1* or Noradrenaline ≤0.1*	Adrenaline >0.1* or Noradrenaline >0.1*
Glasgow Coma Scale	15	13-14	10-12	6-9	<6
Bilirubin (mg/dl)	<1.2	1.2-1.9	2.0-5.9	6.0-11.9	≥12
Creatinine (mg/dl) or Urine output	<1.2	1.2-1.9	2.0-3.4	3.5-4.9 or <500ml/day	≥5.0 or <200ml/day

*µg/kg/min

Each variable is scored between 0 and 4. The SOFA score is the sum of the score of each variable.

APPENDIX VI Combinations of immunotests guiding randomization.

Each patient is screened on two serial time points between two consecutive days for lab findings of fulminant hyper-inflammation or sepsis-associated immunoparalysis. There are 4 different combinations of results that may derive after this screening. Only of these combinations may allow screened patients to be enrolled.

Interpretation	Fulminant hyper-inflammation		Unclassified	Sepsis-associated immunoparalysis
Ferritin (ng/ml)	↑	↑	=	=
HLA-DR (AB/CD14-cell)	=	↓	=	↓

↑ ferritin >4,420 ng/ml;

= ferritin ≤4,420 ng/ml;

↓ HLA-DR on monocytes <5,000 AB/C;

= HLA-DR ≥5,000 AB/C

PERSONALIZED IMMUNOTHERAPY IN SEPSIS: A MULTICENTRE AND MULTINATIONAL, DOUBLE-BLIND, DOUBLE-DUMMY RANDOMIZED CLINICAL TRIAL (THE IMMUNOSEP TRIAL)

APPENDIX VII Study visits

	Study visits																		
Day	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	21	28	90
Obtain ICF	X																		
Study drug		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X			
SOFA score		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Medical history		X																	
Clinical state of infection							X	X	X	X	X	X	X	X	X	X	X		
Survival																		X	X
Vital signs		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Lab tests		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Microbiology		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Blood collection		X	X		X			X								X	X	X	X
Microbiome samples		X	X		X			X								X	X	X	X
Co-administered medication		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Adverse events			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

**PERSONALIZED IMMUNOTHERAPY IN SEPSIS: A MULTICENTRE AND
MULTINATIONAL, DOUBLE-BLIND, DOUBLE-DUMMY RANDOMIZED CLINICAL
TRIAL (THE IMMUNOSEP TRIAL)**

STATISTICAL ANALYSIS PLAN

Authors:

Frank Bloos, Md, PhD

Konstantina Dakou, BSc, MSc

Evangelos J. Giamarellos-Bourboulis, MD, PhD

Miltiades Kyprianou, BSc, Biostatistician

Alexander Vlaar, MD, PhD

Sponsor and CRO:

Hellenic Institute for the Study of Sepsis

17 Laodikeias Str., 11528 Athens, Greece

insepsis@otenet.gr

tel: 0030 210 7480662

EudraCT number: 2020-005768-74

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LIST OF ABBREVIATIONS

AE	adverse event
APACHE	Acute physiology and chronic health evaluation
CCI	Charlson's comorbidity index
CI	confidence interval
HISS	Hellenic Institute for the Study of Sepsis
ITT	Intent-to-treat
MALS	macrophage activation-like syndrome
OR	odds ratio
RCT	randomized clinical trial
ROC	receiver operator characteristics
SD	standard deviation
SOFA	sequential organ failure assessment
TEAE	treatment-emergent adverse events
VAP	ventilator-associated pneumonia

BACKGROUND

The ImmunoSep trial is running since January 2021 in six European countries (Germany, Greece, Italy, the Netherlands, Romania, Switzerland) with the Hellenic Institute for the Study of Sepsis (HISS) as Sponsor. The aim of this randomized controlled trial is to assess whether personalized adjunctive immunotherapy directed against a state of either fulminant hyper-inflammation or immunoparalysis is able to change sepsis outcomes. Patients will be selected by a panel of biomarkers and laboratory findings and will be allocated to placebo or immunotherapy treatment according to their needs. Details have been published in the study protocol¹. The study enrolment will be competitive between the participating study sites.

STUDY DESIGN

This is a prospective multicenter randomized placebo-controlled phase 2 clinical trial in a total number of 24 study sites.

This clinical study falls under Directive 2001/20/EC (Clinical Trials Directive). The study protocol will be submitted for approval to the Ethics Committees (EC) and Institution Review Boards (IRB) as per participating study site regulations. For countries where national authority approval is needed, the protocol was submitted to the central regulatory (e.g. National Ethics Committee of Greece and National Organization for Medicines of Greece). The study was registered under EurdraCT 2020-005768-74 and at ClinTrials.gov (<https://clinicaltrials.gov/study/NCT04990232>).

STUDY POPULATION

The flow of patients through the study will be displayed in a CONSORT diagram. We will report the number of patients screened, met inclusion or exclusion criteria, were enrolled, and were eligible but not enrolled. Reasons for the exclusion of non-included patients will be reported. We will also report the number of patients who were randomized to each group, received the allocated interventions, withdrew informed consent, and had the primary outcome data.

The analysis of the trial is based on the principles of intent-to-treat (ITT) population. ITT is defined as patients randomized to one arm of treatment. Comparators are personalized immunotherapy (consisting of anakinra or interferon

gamma) and placebo. Patients who withdrew informed consent after randomization are removed from the ITT.

PATIENT BASELINE DEMOGRAPHICS

We will present baseline demographics, Acute Physiology and Chronic Health Evaluation (APACHE) II score, Charlson Comorbidity Index (CCI), and baseline Sequential Organ Failure Assessment (SOFA) score as mean and SD, if they follow normal distribution and as median and quartiles if they follow non-linear distribution. Baseline laboratory values will also be presented as means and SD. Baseline data will include frequencies of focus of underlying infection, septic shock (according to Sepsis-3 definition), mechanical ventilation, need of vasopressors, need of continuous renal replacement therapy (CCRT), administered antibiotics and microbiology. In addition, pre-enrolment results of ferritin serum concentrations and of the expression of HLA-DR molecules on CD14/CD45 positive cells will be reported.

ANALYSIS FOR THE STUDY ENDPOINTS

Primary study endpoint

The primary efficacy study endpoint is the comparative difference in the number of patients achieving a 1.4 decrease from baseline to mean total SOFA (Sequential Organ Failure Assessment) score from day 2 until day 9 after randomization

Analysis of the primary endpoint will be done as follows: the mean SOFA score between days 2 and 9 will be calculated by adding the SOFA scores of days 2 to 9 (total of eight values) and dividing by eight. For patients dying before day 9, the missing values will be replaced by 24 which is the maximum of SOFA points a patient may get. For patients discharged home before day 9, the last recorded SOFA point will be carried forward. For patients lost to follow-up, the last recorded SOFA point will be carried forward. Once the mean SOFA score between days 2 and 9 is calculated, the change of the mean SOFA score until day 9 will be calculated by the formula:

$$\Delta\text{SOFA score by day 9} = \text{SOFA score of day 1} - [\text{mean SOFA score of days 2 to 9}]$$

Then patients a binary outcome is created where patients are split into those a) achieving the primary endpoint and who are defined as experiencing ΔSOFA score by

day 9 more than 1.4 points; and b) into those NOT achieving the primary endpoint and who are defined as experiencing Δ SOFA score by day 9 equal to or less than 1.4 points.

For the raw analysis, achievement of 1.4 points decrease in SOFA is compared in the treatment-group with the placebo-group by the Fisher's exact test. The odds ratio (ORs) and 95% confidence intervals (CIs) are calculated.

The analysis of the primary endpoint is further confirmed by one step-wise logistic regression model. For the needs of this model, the achievement of the primary endpoint is entered as dependent variable and the following variables enter as independent variables:

- Baseline APACHE II score as continuous variable
- CCI as continuous variable
- SOFA score as continuous variable
- The study site as binomial variable; this binomial variable is generated after analyzing the frequencies of patients enrolled in each study sites. A total of 31 study sites have participated in the study, so that it is anticipated that up to 8% of enrolled patients can come from one study site. In case of sites providing more than 8% of enrolled patients, patients coming from these sites will be merged in this variable under the name "Patients coming from large sites". All remaining patients will be merged in this variable under the name "Patients coming from other sites".

ORs and 95% CIs are calculated. Any 2-sided p-value less than 0.05 is considered statistically significant.

Secondary endpoints

Achievement of the primary endpoint according to the state of immune activation

The two groups of treatment are compared regarding the achievement of the primary endpoint or not by the Fisher's exact test separately for patients with MALS and for patients with sepsis-induced immunoparalysis. The odds ratio (ORs) and 95% confidence intervals (CIs) are calculated according to Mantel and Haenszel. The ORs for the achievement of primary endpoint for each stratum of immune activation (MALS/sepsis-induced immunoparalysis) are compared by the Breslow-Day and Tarone's test.

In each of states of immune activation (MALS/immunoparalysis), the mean SOFA score is compared only for survivors until day 9 using the Student's t-test. The mean SOFA score is provided by the formula:

[SOFA score day 1 + SOFA score day 2 + SOFA score day 3 + SOFA score day 4 + SOFA score day 5 + SOFA score day 6 + SOFA score day 7 + SOFA score day 8 + SOFA score day 9]/9

Impact of immunotherapy on 28-day mortality

This will be analyzed by plotting of survival curves for patients allocated to placebo treatment and for patients allocated to immunotherapy treated by Kaplan-Meier and by comparisons using Cox regression analysis and reporting the relevant hazard ratios and 95% CIs. Subgroup analysis will be done for patients with MALS and for patients with sepsis-induced immunoparalysis. 28-day mortality is also reported as frequencies and compared by Fisher's Exact test. The odds ratio (ORs) and 95% confidence intervals (CIs) are calculated.

Impact of immunotherapy on 90-day mortality

This will be analyzed by plotting of survival curves for patients allocated to placebo treatment and for patients allocated to immunotherapy treated by Kaplan-Meier and by comparisons using Cox regression analysis. Subgroup analysis will be done for patients with MALS and for patients with sepsis-induced immunoparalysis. 90-day mortality is also reported as frequencies and compared by Fisher's Exact test. The odds ratio (ORs) and 95% confidence intervals (CIs) are calculated.

Change of mean total SOFA score on day 15 of the end of treatment

Analysis as follows: the mean SOFA score between days 2 and 15 will be calculated by adding the SOFA scores of days 2 to 15 (total of 14 values) and dividing by 14. For patients dying before day 15, the missing values will be replaced by 24 which is the maximum of SOFA points a patient may get. For patients discharged home before day 15, the last recorded SOFA point will be carried forward. For patients lost to follow-up, the last recorded SOFA point will be carried forward. Once the mean

SOFA score between days 2 and 15 is calculated, the change of the mean SOFA score until day 9 will be calculated by the formula:

$\Delta\text{SOFA score by day 15} = \text{SOFA score of day 1} - [\text{mean SOFA score of days 2 to 15}]$

Then patients will be split into those a) achieving this secondary endpoint and who are defined as experiencing $\Delta\text{SOFA score by day 15}$ more than 1.4 points; and b) into those NOT achieving the primary endpoint and who are defined as experiencing $\Delta\text{SOFA score by day 15}$ equal to or less than 1.4 points.

The two groups of treatment are compared regarding the achievement of this secondary endpoint or not by the Fisher's exact test. The odds ratio (ORs) and 95% confidence intervals (CIs) are calculated according to Mantel and Haenszel. This secondary endpoint is also analyzed separately for patients with MALS and for patients with sepsis-induced immunoparalysis.

In each of states of immune activation (MALS/immunoparalysis), the mean SOFA score is compared only for survivors until day 15 using the Student's t-test. The mean SOFA score is provided by the formula:

$$[\text{SOFA score day 1} + \text{SOFA score day 2} + \text{SOFA score day 3} + \text{SOFA score day 4} + \text{SOFA score day 5} + \text{SOFA score day 6} + \text{SOFA score day 7} + \text{SOFA score day 8} + \text{SOFA score day 9} + \text{SOFA score day 10} + \text{SOFA score day 11} + \text{SOFA score day 12} + \text{SOFA score day 13} + \text{SOFA score day 14} + \text{SOFA score day 15}] / 15$$

The impact of personalized immunotherapy on the reversal of hyper-inflammation or immunoparalysis

This endpoint is censored at day 15 of the end of treatment and it is defined as follows:

a) for patients with MALS as any at least 15% decrease of the baseline serum ferritin; and b) for patients with sepsis-associated immunoparalysis as restoration of Quantibrite to above 8,000 AB/C with serum ferritin below 4,420 ng/ml.

Patients with MALS are split into those achieving at least 15% of baseline ferritin by day 15 or not. Patients with MALS dying before day 15 are considered as not achieving this decrease of ferritin.

Patients with sepsis-induced immunoparalysis are split into those achieving restoration of immunoparalysis or not by day 15 which is defined as Quantibrite above 8,000 AB/C. Patients with induced immunoparalysis dying before day 15 are considered as not achieving this increase of Quantibrite.

The two groups of treatment are compared regarding the achievement of this secondary endpoint or not by the Fisher's exact test. The odds ratio (ORs) and 95% confidence intervals (CIs) are calculated according to Mantel and Haenszel. This secondary endpoint is also analyzed separately for patients with MALS and for patients with sepsis-induced immunoparalysis.

The impact of personalized immunotherapy on infection resolution

This endpoint is censored at day 15 and patients are divided into those achieving or not achieving resolution of the infection for which they were enrolled. The criteria for resolution of each infection are pre-defined in the study protocol¹. The two groups of treatment are compared regarding the achievement of infection resolution or not by the Fisher's exact test. The odds ratio (ORs) and 95% confidence intervals (CIs) are calculated according to Mantel and Haenszel. Separate analysis using the same tests is done for each type of infection (i.e. primary bacteremia, community-acquired pneumonia, hospital-acquired pneumonia and ventilator-associated pneumonia) and for each stratum of immune activation (MALS/immunoparalysis).

Safety

The incidence of serious and non-serious TEAEs will be compared between the two groups of treatment for each of the immune strata (MALS and sepsis-induced immunoparalysis). Comparisons are done by the Fisher's exact test.

In all analyses of secondary endpoints, any 2-sided p-value less than 0.05 is considered statistically significant.

SUBGROUP ANALYSES

Subgroup analyses for severity

Based on the principles of the SAP defined above, patients are split into subgroups for APACHE II, CCI and SOFA score before start of the study drug. The cut-offs of APACHE II, CCI and SOFA which form the subgroup will be measured after plotting

ROC curves which define their discrimination ability for 28-day mortality for the entire ITT population. From the co-ordinate points of the curves the Youden index is estimated and this is used as a cut-off to transform the subgroups of APACHE II, CCI and SOFA score into binary variables

Achievement of the primary endpoint

The ORs for achievement of the primary endpoint will be calculated separately for according to the cut-off of APACHE II, CCI and SOFA scores before start of the study drug within each of the allocated groups of treatment (placebo/immunotherapy but also MALS and sepsis-induced immunoparalysis). The ORs will be compared by the Breslow-Day's and the Tarone's tests.

28-day mortality

The ORs for 28-day mortality will be calculated separately for according to the cut-off of APACHE II, CCI and SOFA scores before start of the study drug within each of the allocated groups of treatment (placebo/immunotherapy but also MALS and sepsis-induced immunoparalysis). The ORs will be compared by the Breslow-Day's and the Tarone's tests.

90-day mortality

The ORs for 90-day mortality will be calculated separately for according to the cut-off of APACHE II, CCI and SOFA scores before start of the study drug within each of the allocated groups of treatment (placebo/immunotherapy but also MALS and sepsis-induced immunoparalysis). The ORs will be compared by the Breslow-Day's and the Tarone's tests.

Subgroup analyses for efficacy of antibiotics

Since the study design of ImmunoSep enrolled patients with health-care associated infections, it is anticipated that many of enrolled patients suffer from infections by multidrug resistant bacteria and empirically prescribed antibiotics for these patients may not be appropriate. Appropriateness of empirically prescribed antibiotics will be considered as administration of antibiotics at the time of the start of the study drug (placebo/immunotherapy) susceptible to the isolated pathogen according to the antibiogram when the antibiogram became known. Separate analysis

for the primary endpoint will be done for patients receiving appropriate antibiotics and for patients receiving non-appropriate antibiotics.

In all subgroup analyses, any p-value less than 0.05 is considered statistically significant.

EXPLORATORY ENDPOINT

ImmunoSep is considered as a major landscape to define new endpoints for RCTs in sepsis management. To this end, ROCs will be plotted comparing APACHE II score, CCI, SOFA score of day 1, change of SOFA score by day 9 and changes of SOFA score by day 15 for the prediction performance for death after 28 days and after 90 days. This analysis will be done for the total of enrolled patients, for patients allocated to placebo treatment and for patients with MALS/sepsis-induced immunoparalysis. The aim of the analysis will be to define a cut-off of each of the studied variables to be used in the future as endpoints for trials in sepsis.

REFERENCE

1. Kotsaki A, Pickkers P, Bauer M, Calandra T, Lupse M, Wiersinga WJ, et al. ImmunoSep (personalised immunotherapy in sepsis) international double-blind, double-dummy, placebo-controlled randomized clinical trial: study protocol. *BMJ Open* 2022; 12: e067251